

UNIVERSITY OF GREENWICH
FACULTY OF ENGINEERING AND SCIENCE
SCHOOL OF COMPUTING AND MATHEMATICAL
SCIENCES

**Social Media Intelligence (SOCMINT)
Aggregation: Deriving Pattern of Life from
Fragmented Digital Traces**

URL: <https://github.com/Jorgearteaga83/PoL-Analysis-OSINT>

Jorge Arteaga Gomez
(1278829)

Supervisor: Anatolij Bezemskij

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Abstract

This project investigates the application of Open-Source Intelligence (OSINT) to synthesize fragmented digital footprints into cohesive Pattern of Life (PoL) profiles, a phenomenon known as the "Mosaic Effect." To empirically demonstrate this, a custom, offline analytical dashboard was developed using Python. The system ingests decoupled, scraped social media data and applies programmatic heuristics, including VADER sentiment analysis, spatial categorization, and temporal windowing to identify behavioral anomalies such as routine disruptions and travel patterns. To strictly adhere to GDPR regulations and ethical academic standards, the research was conducted entirely offline using synthetic datasets, mitigating the risks associated with live human subject data. The results successfully prove that isolated, seemingly benign data points can be programmatically aggregated to reveal sensitive, actionable intelligence about a target's lifestyle, associations, and vulnerabilities.

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 1. To comment code
 2. Debugging and testing
 3. Assisting architecture design

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Contents

Abstract	ii
Declaration of AI Use	iii
Acknowledgements	iv
Table of Contents	vii
List of Figures	viii
1 Introduction	1
1.1 Background and Context	1
1.2 Pattern of Life (PoL) Analysis in OSINT	2
1.3 Security-Oriented Sentiment Analysis in OSINT	3
1.4 Problem Statement	3
1.5 Aim and Objectives	4
1.6 Research Questions	5
1.7 Scope and Limitations	5
1.8 Report Roadmap	5
2 Literature Review	6
2.1 Conceptualizing OSINT and Social-Media Intelligence	6
2.2 OSINT Structures and Cyber Threat Intelligence Pipes	7
2.3 Digital Forensics and Social Media	7
2.4 Multimedia and Image Forensics within Social Media	8
2.5 Inference: Digital Traces, Mobile Data, and Pattern of Life	9
2.6 Social Media Data Leakage, Incidental Disclosure, and Privacy	10
2.7 Sentiment Analysis and Security-Oriented Social Media Analysis	11
2.8 Synthesis and Conceptual Framework	12
2.9 Literature Synthesis and Comparison	13
2.10 Product Review	13
3 Requirements Analysis and Design	14
3.1 Introduction	14
3.2 MoSCoW Prioritization	14
3.3 Data Acquisition and Collection Pipeline	15
3.3.1 Synthetic Dataset Design Parameters	17
3.4 Temporal Windowing and Filtering Strategy	18
3.5 Data Normalization and Analytical Representation	18

3.6	Pattern of Life Operationalization Through Temporal Analytics	19
3.7	Technique of Sentiment Analysis and Competitiveness	19
3.8	Image Leakage/Metadata Risk Analysis	20
3.9	Graphical User Interface as an Interpretability Layer	20
3.10	Activities of evaluation and validation strategy	21
3.11	Summary	22
4	Implementation	23
4.1	Code Repository Outline	23
4.2	Data Sources and Acquisition	24
4.2.1	File-Based Dataset Mode	24
4.2.2	Social Network Analysis (SNA) and Automated Reporting	25
4.2.3	Synthetic Account Generation & Variable Control	26
4.3	Data Preprocessing and Normalization	28
4.4	Temporal Analysis & Pattern of Life Extraction	29
4.4.1	Temporal Windowing	29
4.4.2	Temporal Pattern Metrics	30
4.5	Sentiment Analysis Implementation	30
4.6	Pattern of Life Modelling	31
4.7	Graphical User Interface Design	32
4.8	Reliability, Error Handling, and Robustness	32
4.9	Chapter Summary	33
5	Results and Evaluation	34
5.1	Results	34
5.2	Discussion of Analytical Results	37
5.2.1	Temporal Behavior and Routine Identification	37
5.2.2	Sentiment Trends and Behavioral Context	37
5.2.3	Image Usage and Leakage Indicators	38
5.3	Pattern of Life Interpretation	38
5.4	System Evaluation against Objectives	39
6	Legal, Social, Ethical and Professional Issues	40
6.1	Introduction	40
6.2	Data Sources, Target Accounts, and Ethical Constraints	40
6.3	Methodological Safeguards	41
6.4	Ethical Reflections and Legal Compliance (UK DPA 2018 / GDPR)	41
6.4.1	Limitations of the Study	42
7	Conclusion	43
7.1	Summary of Outcomes and Aims	43
7.2	Evaluation of Research Questions (RQs)	43
7.3	Synthesis of the 'Mosaic Effect'	44
7.4	Technical Limitations	44
7.5	Future Work	44
7.6	Final Remarks	44
	Bibliography	45

A	Project Resources and Documentation	51
A.1	Project Proposal	51
A.2	Source Code (SOCMINT Tool)	51
A.3	Full System Documentation	51

List of Figures

2.1	Mosaic effect bringing several distinct fields together to form a Pattern of Life.	12
3.1	UML Diagram of System Architecture	15
3.2	Flowchart of Data Pipeline	16
3.3	Out of Scope: Dataset generation, In Scope: Ingestion and Profiling . . .	17
3.4	GUI Wireframe layout	21
4.1	Social Network Analysis graph (SNA).	25
4.2	A view of the 6 synthetic Instagram profiles for the SOCMINT analysis. .	27
4.3	A technical comparison demonstrating platform metadata stripping. The raw image file (left) contains geospatial and device artifacts, whereas the platform-processed equivalent (right) has all identifying headers removed.	28
4.4	The primary OSINT analysis dashboard.	32
4.5	High-level data flow architecture.	33
5.1	Temporal Analysis per day tab.	34
5.2	Temporal Analysis per hour.	35
5.3	Post per weekday.	35
5.4	Pattern of Life heatmap.	36
5.5	Daily sentiment trend aggregation.	36

Chapter 1

Introduction

1.1 Background and Context

Contemporary online life is characterized by the continuous generation of unintentional data traces, often referred to as “digital exhaust”. Smartphones, social media, and cloud-based services continuously monitor locations, timestamps, and communications. This data can be measured to understand the movement and habits of humans. Of particular concern is social media, which encourages **self**-disclosure of personal information. Recent empirical studies demonstrate that individuals routinely disclose features such as their city of residence, work, and interpersonal connections, often propelled by algorithmic amplification and engagement metrics (Sykownik et al., 2022[44]; Buty, 2022)[7].

While a single piece of data may not be threatening, when these traces are multiplied over time, they form a mosaic. An adversary can compile these fragments to recreate a detailed user profile, bypassing privacy controls. This is facilitated by the “privacy paradox,” where users systematically underestimate the long-term risks of data aggregation in exchange for short-term social benefits (Jiang, 2024)[20]. Furthermore, incidental **disclosures**—such as background objects in images or secondary leakage from tagged friends—exacerbates this exposure (Zamoff and Bellwood, 2022[52]; Kutschera et al., 2024)[23]. The “privacy calculus” governing these decisions often fails on social platforms due to observational learning and algorithmic curation (Xu & Wang, 2023[49]; Li and Ye, 2025[25]; Leung, 2023)[24].

Open-Source Intelligence (OSINT) has consequently become a vital tool in cybersecurity and threat intelligence. While current OSINT pipelines utilize AI and scraping to monitor threat actors and campaigns at a macro level (Babenko et al., 2025[5]; Alevisos & Dekker[54], 2024[54]; Cerruto et al., 2022)[8], minimal effort has been made to demonstrate how these tools can be used to reconstruct an individual’s Pattern of Life (PoL). This project addresses this gap by investigating how publicly accessible social media traces (timestamps, EXIF metadata, and text) can be aggregated to infer a person’s habitual routines, locations, and emotional states.

1.2 Pattern of Life (PoL) Analysis in OSINT

Pattern of Life (PoL) analysis refers to the methodical replication of recurrent patterns of behavior, time, and space of an entity or phenomenon over time to specify a degree of normal activity. The PoL analysis in the intelligence and security context has a long history of use in war intelligence, geospatial intelligence (GEOINT), and mobility analysis to infer intent, future behavior, and anomalies indicative of risk.

Pattern of Life analysis within the context of Open-Source Intelligence is built exclusively on publicly available digital footprints, particularly those generated during day-to-day social media interactions. Posting time stamps, geolocation artefacts in multimedia, textual data, and metadata are such traces. This information can be organized in systematic, time-based monitoring, and the patterns of the routine can be categorized with ease without necessarily being provided with special authority or coming under scrutiny.

Intelligence practitioners tend to conceptualize the Pattern of Life on three principal analytical planes: Spatial, Temporal, and Behavioral/Digital. These dimensions all determine the place, time, and form in which a person behaves or presents himself or herself in an online environment. Table 1.1 presents the structured classification of these dimensions as applied in Pattern of Life reconstruction using OSINT.

Table 1.1: Pattern of Life Classification Dimensions in OSINT

Dimension	Analytical Focus	OSINT Data Sources	Example Inferences
Spatial	Geographic regularity	EXIF GPS metadata, geotags, landmarks in images	Home location, workplace, frequent venues
Temporal	Time-based routines	Post timestamps, upload frequency	Sleep cycles, work hours, travel periods
Behavioral / Digital	Expressive and interaction patterns	Text content, hashtags, media type, sentiment	Emotional state, lifestyle indicators

The spatial aspect enables the determination of the location of movements and time that people spend most of their time, which are prevalent in residential, occupational, and recreation trends. The time factor is associated with routine and cycle of online life, to be capable of deducing the habits of everyday routine and window of availability, long absence. The expressive patterns captured in the behavioral or digital dimension include sentiment trends, content preferences, and modes of interaction that may be utilized in defining the states of emotion or the progression of lifestyle.

Pattern of life analysis includes a significant value other than the baseline, which is anomaly detection. Deviations of patterns; instant paralysis, unexplainable spatial mobility, or some sharp turnaround in mentality is normally more instructive than ritual action. These pattern breaks are typical indications of key events in the practice of intelligence to include travel, stress, change of operation, or deliberate evasion. The latter is the principle of the analytical design of the proposed system that points out the visibility of the routine structures to make intuitively visible deviations.

1.3 Security-Oriented Sentiment Analysis in OSINT

Sentiment analysis is the third pillar of this project’s analysis. Sentiment and opinion mining are emerging techniques in OSINT processing that aim to identify attitudes, emotional states, and intent to act by analyzing publicly available text-based information. Radicalization, crisis monitoring, and threat perception have been the areas where security-conscious sentiment analysis has been applied (Liu & Shi, 2022)[26].

The scale of data generated on social platforms necessitates automated approaches; machine learning algorithms are now routinely deployed to analyze the sentiment of posts and comments, allowing analysts to extract meaningful behavioral perceptions and trends from vast volumes of unstructured public data (Leung, 2023)[24].

In the context of this dissertation, unlike detecting threats and intent to harm, the degree of emotional and attitudinal tendencies is assessed through sentiment analysis of publicly available posts by a subject on social networks. It uses a lexicon-based sentiment analysis model to process textual data, generate an aggregate emotional profile over time, and categorize posts into positive, neutral, and negative categories.

Sentiment trends provide an added dimension of behavior (when combined with Pattern of Life analysis) by enhancing the association between emotional expression, time routines, and local activity. Sentiment outputs will be visualized on well-marked time axes and accompanied by brief interpretive summaries, which will provide context for the emotional tone of each account under analysis. This approach brings an analytical focus while maintaining openness to constraints and the probabilistic nature of sentiment inferences.

1.4 Problem Statement

The existing research on OSINT, social media analytics, and digital forensics demonstrates that, technically, one can draw conclusions about personal habits, places, routines, and attitudes from publicly available social media information. The automated processes of scraping, correlation, and analytical technologies have been automated; the user has never valued the problems of privacy and data aggregation.

However, most of the existing literature and equipment operate either at a macro level (i.e., large population, campaigns, or communities of threats) or target professional researchers and intelligence practitioners. Minimal effort has been made to demonstrate, in a managed and moral manner, how individual users can be profiled solely from their well-recognized digital trail.

The central question addressed by this study is as follows:

To what extent does a post-acquisition OSINT analysis engine ingesting standardized datasets of publicly available social media traces form a rich Pattern of Life, place profile, and sentiment profile of an individual, thereby demonstrating how dangerous data integration is on the individual level?

1.5 Aim and Objectives

According to the project proposal adopted, the common goal of the thesis is:

To create, implement, and test a Python-based OSINT application that ingests, normalizes, and combines pre-collected, publicly shared social media datasets (images, text, and timestamps) in order to create a more composite pattern of life and sentiment profile to demonstrate a high, but extremely underrated, confidence threat.

This objective is operationalized under the following objectives:

- Review the available literature on data leakage on social media, application of OSINT tools, pattern of life analysis, and what legal/ethical concerns are associated with scraping publicly available data.
- Determine the most effective solutions for EXIF metadata extraction, temporal analysis, and sentiment analysis within OSINT.
- Denote a modular structure that entails Data Ingestion, EXIF Analyzer, Timestamp Analyzer, Sentiment Analyzer, and Visualizer modules, designed to operate exclusively on post-acquisition offline datasets.
- Decide on the data model and storage formats for processing collected content, specifically utilizing in-memory Pandas DataFrames and standardized CSV exports for persistence.
- Install EXIF analysis to extract geolocation and other available metadata in images that have been downloaded.
- Develop a post-acquisition ingestion module capable of securely processing standard offline datasets (e.g., CSV, XLSX). This ensures strict Operational Security (OPSEC) by preventing external API calls that could alert target platforms, while maintaining analytical repeatability. Additionally, implement a Social Network Analysis (SNA) module utilizing graph theory to map associated entities and highlight secondary data leakage through tagged users.
- Use timestamp analysis to get PoL heatmaps.
- Perform sentiment analysis of text in social media with an adequate model (e.g., VADER).
- Identify interested parties and implement the tool upon their social profiles.
- Strongly question the technical, legal, and ethical concerns regarding the building of such an instrument.
- Take into consideration the implications of user education, platform design, and privacy regulation.

1.6 Research Questions

Consistent with the aim and objectives, the thesis will respond to the following research questions (RQs):

- **RQ1 - Data availability:** How effective is the selected social media platform in enabling the extraction of EXIF metadata, timestamps, and textual data from publicly accessible profiles without administrative or privileged access?
- **RQ2 - Pattern of life reconstruction:** How well can the pattern of posting times and geotagged images be utilized to estimate the routine of a participant (i.e., home time, working hours, travelling patterns)?
- **RQ3 - Sentiment profiling:** What kind of attitudinal or emotional patterns can be inferred based on sentiment analysis on the open posts of a participant, and how stable are they over time?
- **RQ4 - Risk perception:** How does exposure to aggregated patterns of life and sentiment profiles influence perceptions of social media privacy and data leakage risks?

1.7 Scope and Limitations

This project deliberately restricts its scope to ensure it remains ethical and technically viable:

- **Single main platform:** It is a prototype devoted to one mainstream social media platform to coordinate technical complexity and respect rate limitations and anti-scraping measures.
- **Only publicly available information:** The tool will not attempt to authenticate, break access controls, or perform other technical exploits. It will only process publicly available data, which is consistent with common OSINT practice and legal recommendations.
- **None of the real-world targeting:** The tool will be an informational artefact; it will neither be published nor implemented against unconsenting real-world targets.

It is also anticipated to be technologically limited, as platforms completely remove the EXIF information of uploaded photos (resulting in weak geolocation signals), and the sentiment analysis of short and context-poor posts is not a flawless task yet. Such constraints will be seen as experiment results rather than project failures.

1.8 Report Roadmap

Chapter 2 reviews the literature surrounding OSINT and PoL, and compares existing tools. Chapter 3 outlines the requirements analysis using MoSCoW prioritization and details the system design. Chapter 4 covers the implementation of the Python-based tool. Chapter 5 presents the results and evaluates the system's effectiveness. Chapter 6 critically reflects on the legal, social, ethical, and professional issues. Finally, Chapter 7 concludes the project and outlines future work.

Chapter 2

Literature Review

2.1 Conceptualizing OSINT and Social-Media Intelligence

Open-source intelligence (OSINT) has evolved to be a distinct practice to what some authors today call a fifth intelligence discipline, on par with signals, human, and imagery intelligence (Yadav et al., 2023)[50]. Van Puyvelde and Tabarez[45] reveal that the increasing volume of digital traces, including social media, open government data, and business sites, has put OSINT at the core of contemporary intelligence practice, and harder to manage and justify by its bulk and diversity (Van Puyvelde and Tabarez, 2025)[45]. The transformation will make a difference in the ongoing project since this project entirely relies on unfiltered and unrestricted social media content.

Within the context of cybersecurity, OSINT no longer occurs via manual search but rather is applied in automated pipelines. To describe the way the OSINT system can be scaled and used for a defensive effort, Babenko et al. (2025)[5] present an automated system of OSINT to detect the exposed digital assets of an organization and calculate the cyber risk through the synthesis of web crawling, asset fingerprinting, and risk scoring. The preference for automation is also evident in the case of cyber threat intelligence (CTI). Alevizos & Dekker (2024)[54] design an AI-enhanced CTI pipeline, which uses OSINT feeds, enrichment modules, and machine learning classifiers to give priority to compromising indicators, providing value in the structured acquisition, normalization, and scoring of the resultant public data.

The other facet of OSINT that has attracted the attention of researchers is the educative and knowledge organization facet. Herrera-Cubides et al.[18] chart the topical clusters (cybercrime, terrorism, social media monitoring) based on the Senti-o-metric analysis of open-access OSINT literature and instructional material, showing that OSINT research has developed at a rapid rate since 2010 (Herrera-Cubides et al., 2020)[18]. OSINT's multidisciplinary character, comprising computer science, international security, law, and communication studies, is reflected in its practice and in the mixed technical-legal nature of this dissertation.

2.2 OSINT Structures and Cyber Threat Intelligence Pipes

According to some authors, effective OSINT collection in the direction of security should seek precise formations rather than arbitrary searches. Sanchez et al.[38] provide an overview of cyber intelligence frameworks and explain how AI may be introduced at every point of the intelligence cycle, beginning with the collection process and concluding with dissemination and analysis (Sánchez del Monte & Hernández-Álvarez, 2023)[38]. They offer interoperability of data models and traceability of analysis actions, which are concerns that directly regard the auditability of an OSINT tool that assembles data on social media into pattern of life profiles.

CTI generation workflows also rely on automation. Sun et al. (2021)[43] note that there is an automatic CTI generation mechanism, which feeds on OSINT feeds, runs them through a natural language processing engine to extract entities (e.g., malware names and vulnerabilities), and creates formatted threat reports. Preuveneers et al. (2021)[37] proposed a scheme of sharing trained machine learning models as indicators of compromise in the sense that trained models based on OSINT-generated features can be reusable CTI artefacts across organizations.

Defensive deception has also been studied with respect to OSINT. In an OSINT-based moving target defense plan presented by Seo and Kim (2021)[39], the publicly available information about the infrastructure of a particular organization is altered by exploiting false decoys, thus skewing foreign reconnaissance and making attacks more expensive. At the same time, Yuan et al. (2024)[51] construct a toolkit that runs an OSINT pipeline and lets a large language model invoke scraping and enrichment tools when performing CTI analysis, demonstrating how OSINT processes, such as entity resolution and indicator correlation, can be coordinated with tool-augmented LLMs.

Together, these works demonstrate that modern OSINT is characterized by automated, modular pipelines, distinct data processing steps, and an increased reliance on AI. Those trends follow the architecture of this project—which separates scraping, metadata extraction, timestamp analysis, and sentiment analysis into individual modules—serving as a small-scale example of such a pipeline for the analysis of privacy-sensitive patterns of life.

2.3 Digital Forensics and Social Media

While OSINT suggests an offensive or intelligence-based perspective, digital forensics seeks to provide rigorous responses to evidence such as attribution, timespan, and corroboration. Social media artefacts as significant aspects of digital investigations are highlighted in recent literature. According to David, Morris, and Appleby-Thomas (2020)[11], a two-step model of social network investigations is proposed in which investigators first recreate social actions and the creation of technical artefacts, and afterward locate such artefacts within the context of the case investigation questions. They argue that investigators must be aware of both the technical procedures of the platform and the social interpretation of posts and interactions.

The use of AI has helped in such analysis. Bokolo & Liu (2024)[6] conduct a systematic review of artificial intelligence tools that can be used in the field of social media forensics, dividing them into categories of machine learning applied to questionable ac-

count detection, cross-platform and multi-media forgery detection, and behavioral profiling. However, they remark that most solutions presuppose publicly and freely available high-quality information, but do not fully consider the ethical or privacy consequences—a factor this dissertation attempts to preempt.

Blockchain is also beginning to appear as an integrity system for evidence. Shahbazi and Byun (2022)[40] present a forensic analysis system based on NLP with an online social network where a blockchain stores the hash of the extracted evidence, enabling the verification of peripheral evidence with the assistance of chat logs and posts. Their work illustrates how text mining pipelines, such as the sentiment component developed in this project, may be integrated into more formal evidential chains. On a broader scale, Nath et al. (2024)[31] argue that digital forensics workflows should explicitly illustrate the chain of custody, preservation, and acquisition decisions, allowing courts to establish the reliability of digital evidence.

When it comes to specific artefacts of social media, Gupta et al. (2023)[17] provide a comprehensive view of artefact recovery for well-known social media applications, including the recovery of chats, geotags, and deleted messages on client devices and via provider APIs. Their literature analysis shows that despite a platform removing observable identifiers, such as EXIF geotags, secondary artefacts like cached thumbnails or synchronized app databases may still reveal patterns of location and time. Complementing this, Kazare et al. (2024)[21] comment on the forensic procedures relating to social media use in cybercrime cases, emphasizing the necessity to acquire platform logs in addition to device-side evidence to rebuild the sequence of posting and interactions.

Furthermore, Malik (2024)[28] discusses the evolving environment of digital forensic models, indicating how traditional linear models are being replaced with iterative and feedback models better suited to keep up with existing continuous data streams and cloud-stored data, such as social media logs. Their argument mirrors this project’s scrape-and-analyze process, which proceeds through iterative cycles to refine a pattern-of-life profile over time rather than depending on a single scrape.

2.4 Multimedia and Image Forensics within Social Media

The largest confidentiality risk examined by this project lies in images and videos. Pasquini et al. (2021)[35] provide an overview of media forensics on social media platforms, encompassing approaches to identifying source cameras, detecting manipulation, and tracing provenance. They note that the transcoding and metadata stripping inherent to social media platforms can significantly affect the forensic workflow.

Deep learning is now heavily implemented in image forensics research. Shi et al. (2023)[42] surveyed deep learning-based image forensics algorithms, detailing forgery localization, GAN image detectors, and camera model typification. They note that EXIF metadata may be useful to augment device and timestamp verification but is often missing or tampered with on social sites, forcing investigators to increasingly rely on content-based features. Another focal point is fake image detection and mitigation on social media, encompassing adversarial examples and the speed at which fabricated content spreads across networks (Sharma et al., 2023)[41].

Concerning privacy, DeHart et al. (2020)[12] prove that the images users leave on social networking sites tend to reveal sensitive situational details even when identifying

characteristics are obscured. Based on their examination of Twitter posts, they conclude that individuals heavily underestimate the extent to which faces, backgrounds, and contextual information reveal personal details, describing the sharing of social media images as a scourge of visual privacy. This finding forms the foundational design choice for the present project: treating every image not only as a potential source of explicit data (e.g., EXIF GPS) but also as an approximation of location and activities based on aggregated averages.

2.5 Inference: Digital Traces, Mobile Data, and Pattern of Life

Digital forensics researchers increasingly recognize that an individual artifact holds little evidential value unless analyzed within the context of a larger pattern of action. Okmi et al. (2023a)[33] present an overview of mobile phone data and how call logs, application logs, and location traces generate patterns of communication, mobility, and social organization for urban planning and criminology. Building on this, they survey how aggregated traces are used in criminal investigations to prioritize suspects, assess opportunity, and provide corroboration (Okmi et al., 2023b)[34].

Table 2.1: Pattern of Life Analytical Objectives

Analytical Goal	Description	Example Inference
Baseline Modelling	Identify normal behavior	Daily activity cycles
Anomaly Detection	Detect deviations	Unusual travel
Behavioral Profiling	Characterize routines	Lifestyle patterns
Risk Exposure	Assess vulnerability	Predictable absence

The present study focuses on baseline modelling and anomaly visibility as an educational red-team exercise, rather than operational intelligence targeting.

Pattern of lifestyle analysis is particularly evident in location-centric applications. Nunes et al. (2024)[32] analyzed six Android fitness applications, demonstrating that workout histories and GPS traces could be extracted to reproduce the regular routes, times, and speeds of users. Although based on sports apps rather than social media, the core concept remains: repetitive events with timestamps combine to form a highly informative behavioral signature. The autoPoD Mobile framework, a synthetic mobile forensics training dataset generator, further demonstrates how behaviorally plausible digital traces (messages, app usage, movements) can be procedurally generated to mimic human routines (Michel et al., 2022)[30].

IoT forensics extends this notion to the broader sensor ecosystem. Ahmed et al. (2024)[1] map the ontology of IoT forensics, proving how domestic devices, wearables, and embedded systems continuously leave digital traces that can reconstruct the routines and presence patterns of individuals. Although this project concerns social media platforms

rather than IoT systems, the two share a critical similarity: higher-level activity must be inferred from large volumes of low-level traces, and the aggregation of such traces over time raises severe privacy concerns.

These findings validate the project’s decision to develop a 24/7 activity heatmap using post timestamps. The tool aims to translate raw digital traces into interpretable behavioral features by aggregating posting times into distinct diurnal and weekly patterns, an approach strongly advocated in current digital forensics literature.

2.6 Social Media Data Leakage, Incidental Disclosure, and Privacy

Empirical evidence is abundant regarding how social media users unknowingly present personal data, and the likelihood of that data being misapplied. In a systematic review of privacy and disclosure models, Wang and Yu (2021)[47] report the presence of shared constructs—privacy concerns, perceived control, and social norms—and how their interactions drive disclosure behaviors. Their assessment lends credence to the “privacy calculus” theory, wherein users weigh the perceived benefits of sharing against abstract risks, often sharing in a way that underestimates the threat of data aggregation.

Koohang et al. (2021)[22] investigated privacy and security concerns among social media users, finding that despite users reporting moderate to high concern levels, they exhibited a low level of technical awareness regarding specific risks (e.g., metadata leakage). This continuous data leakage is frequently a byproduct of poor user interface design and complex privacy controls. As Alemany, del Val, and García-Fornes (2022)[3] note, users consistently struggle with privacy decision-making mechanisms, leading to a gap between their intended privacy preferences and their actual disclosure behavior. This systemic oversharing provides the foundational data required for OSINT PoL profiling. Similarly, Liu et al. (2022)[27] showed that convenience, social connectedness, and platform design features often override privacy concerns.

Table 2.2: Classification of Social Media Data Leakage

Leakage Type	Description	Example
Primary Leakage	Direct self-disclosure	Posting a location
Secondary Leakage	Disclosure by others	Tagged photos
Incidental Leakage	Background information	Objects in images
Aggregated Leakage	Cross-post correlation	Routine reconstruction

Beyond self-reported attitudes, incidental and secondary leakage are significant problems. Kutschera et al. (2024)[23] explore incidental data disclosure, where sensitive information in the background of a photo is accidentally revealed. They found that one in five survey participants would willingly post an image that the authors considered to be privacy-damaging. The scale of this issue at a network level is further described by Ceruto et al. (2022)[8], who show how sensitive features can be forecasted using community patterns even when explicit identifiers are removed. Vgena et al. (2022)[46] demonstrate that while individual profile fields (e.g., occupation, relationship status) seem harmless, combining them greatly increases the possibility of a privacy breach, especially when data is cross-referenced between platforms.

Furthermore, Zhu et al. (2022)[53] investigated TikTok, uncovering how interactive features like duets, stitches, and algorithmic suggestions prompt harmful disclosure behaviors, especially when coupled with gamified interaction signals. The consensus across this literature (DeHart et al., 2020[12]; Kutschera et al., 2024[23]; Vgena et al., 2022)[46] is that the primary privacy risk stems not from isolated data points, but from the “mosaic effect” created by aggregating several posts, pictures, and metadata factors over a user’s lifetime. Demonstrating this mosaic effect is the primary goal of the proposed OSINT tool.

2.7 Sentiment Analysis and Security-Oriented Social Media Analysis

Sentiment analysis is a well-established technique in social media research and has recently been connected to security and privacy. In a review of sentiment analysis in social media security, Gupta and Kumar (2025)[16] show how affective signals derived from posts are used to detect cyberbullying, monitor radicalization patterns, and gauge reactions to incidents. They argue that combining sentiment, temporal, and network features models user behavior more effectively than content alone.

Md Suhaimin et al. (2023)[29] list sentiment analysis tools utilized in public security research, including lexicon-based approaches, classical machine learning classifiers, and deep neural networks. They note that lexicon-rule tools, such as VADER, remain highly effective for processing short, informal social media messages since they handle emojis and slang well without requiring massive training datasets. This is supported by Xu et al. (2022)[48], who synthesized the literature on social media sentiment analysis, highlighting the ongoing challenges of sarcasm, domain adaptation, and multilingual data.

Alam et al. (2025)[2] explain the use of sentiment aggregation in policy decision-making, allowing authorities to monitor citizen responses to interventions. However, they warn that non-representative samples and black-box models can misinform policymakers. This warning is highly relevant to this project; while a target’s emotional state can be approximated using sentiment scores, it remains a probabilistic inference subject to model bias and situational vagueness.

2.8 Synthesis and Conceptual Framework

The consensus across OSINT, digital forensics, privacy research, and sentiment analysis literature is clear: seemingly banal digital traces become highly sensitive intelligence when assembled in large quantities, cross-linked, and interpreted over time. State-of-the-art OSINT literature demonstrates that open-access data can be processed through modular, AI-supported pipelines to generate structured intelligence products (Babenko et al., 2025[5]; Yuan et al., 2024[51]; Alevizos & Dekker, 2024)[54]. Digital forensics literature corroborates that these pipelines can reconstruct high-fidelity behavioral narratives using device logs and internet artifacts (David et al., 2020[11]; Okmi et al., 2023)[33].

At the same time, the existing sources about privacy characterize a steady discrepancy between the expressed concerns and the behavior of the users, particularly regarding incidental disclosures and the compounding visibility of network sharing (Kutschera et al., 2024[23]; Vgena et al., 2022)[46]. Visual privacy studies emphasize that images contain vast amounts of both explicit metadata and implicit contextual knowledge (Pasquini et al., 2021)[35]. Finally, sentiment analysis confirms that text residues can predict attitudinal states, which, when integrated with temporal traits, provide a comprehensive picture of behavioral routines (Md Suhaimin et al., 2023)[29].

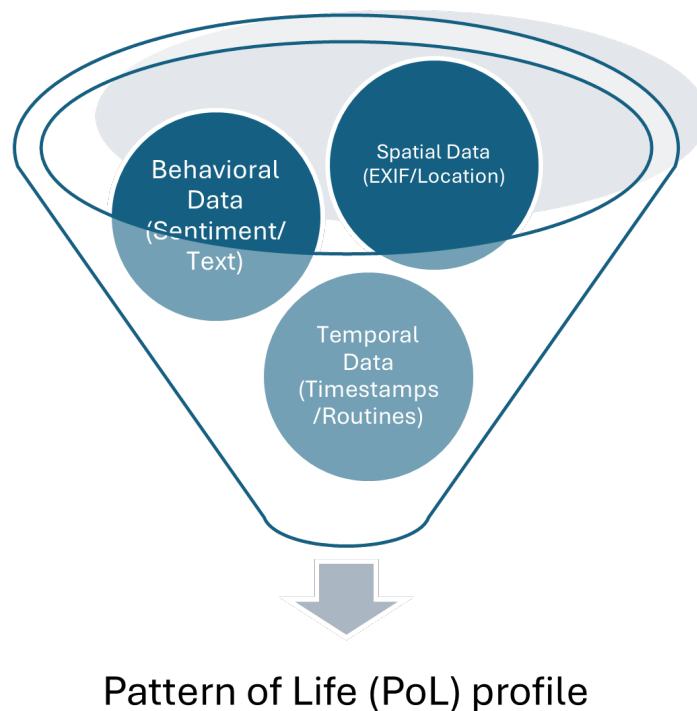


Figure 2.1: Mosaic effect bringing several distinct fields together to form a Pattern of Life.

2.9 Literature Synthesis and Comparison

Table 2.3: Comparison of Key Literature Themes

Author(s) & Year	OSINT Automation	Digital Forensics / EXIF	Sentiment Analysis	Privacy Paradox / PoL
Babenko et al. (2025)[5]	✓	×	×	×
David et al. (2020)[11]	✓	×	×	×
DeHart et al. (2020)[12]	×	✓	×	✓
Jiang (2024)[20]	×	×	×	✓
Md Suhaimin et al. (2023)[29]	×	×	✓	×
Okmi et al. (2023)[33]	×	✓	×	✓

2.10 Product Review

To justify the development of this PoL analysis tool, it is necessary to evaluate existing systems in the problem domain. Industry-standard OSINT tools, such as Maltego and SpiderFoot, excel at infrastructure mapping and threat intelligence gathering. However, they are primarily designed for offensive security professionals targeting enterprise networks or broad cybercrime campaigns. They lack dedicated, automated modules for synthesizing individual, temporal ‘Pattern of Life’ heatmaps combined with sentiment analysis. By building a custom Python-based tool focusing specifically on incidental disclosure (EXIF data) and routine mapping (timestamps), this project fills a distinct gap in demonstrating personal privacy risks rather than enterprise vulnerabilities.

Chapter 3

Requirements Analysis and Design

3.1 Introduction

This chapter outlines the system requirements and architectural design for the PoL OS-INT tool.

3.2 MoSCoW Prioritization

To define the system specifications clearly, the MoSCoW prioritization method was applied:

Must Have:

- Python 3.x environment.
- Post-acquisition ingestion module to securely process offline datasets (CSV/XLSX) exported from Apify.
- Timestamp analyzer to group posting frequencies (time of day, day of week).
- Graphical User Interface (GUI) to display the outputs.

Should Have:

- Sentiment Analysis module utilizing VADER to assign emotional polarity scores to captions.
- Automated generation of temporal heatmaps via matplotlib.

Could Have:

- Social Network Analysis (SNA) using graph theory (NetworkX) to map tagged users and secondary leakage.
- EXIF analysis to extract geolocation metadata from images.

Won't Have:

- Live API scraping directly within the tool (to maintain OPSEC and avoid platform rate limits)..

3.3 Data Acquisition and Collection Pipeline

The initial working stage of the method is data acquisition. To mitigate the inherent volatility of modern social media platforms, raw data acquisition is strategically decoupled from the primary analysis tool. Platforms such as Instagram operate as complex Single Page Applications (SPAs) that employ frequent architectural changes, dynamic data loading, and aggressive anti-scraping countermeasures. Developing a custom-built scraping module within the tool would create a fragile dependency, highly susceptible to breaking whenever the platform updates its interface or API restrictions. Therefore, to ensure the long-term reliability and viability of the project, data extraction is delegated to an established, specialized third-party service (Apify). This service securely navigates platform restrictions to extract the target's digital footprint, exporting it as standardized offline datasets (e.g., .csv files). This architectural choice shields the analytical pipeline from upstream data-gathering failures, allowing the system to focus exclusively on robust post-acquisition normalization and Pattern of Life inference.

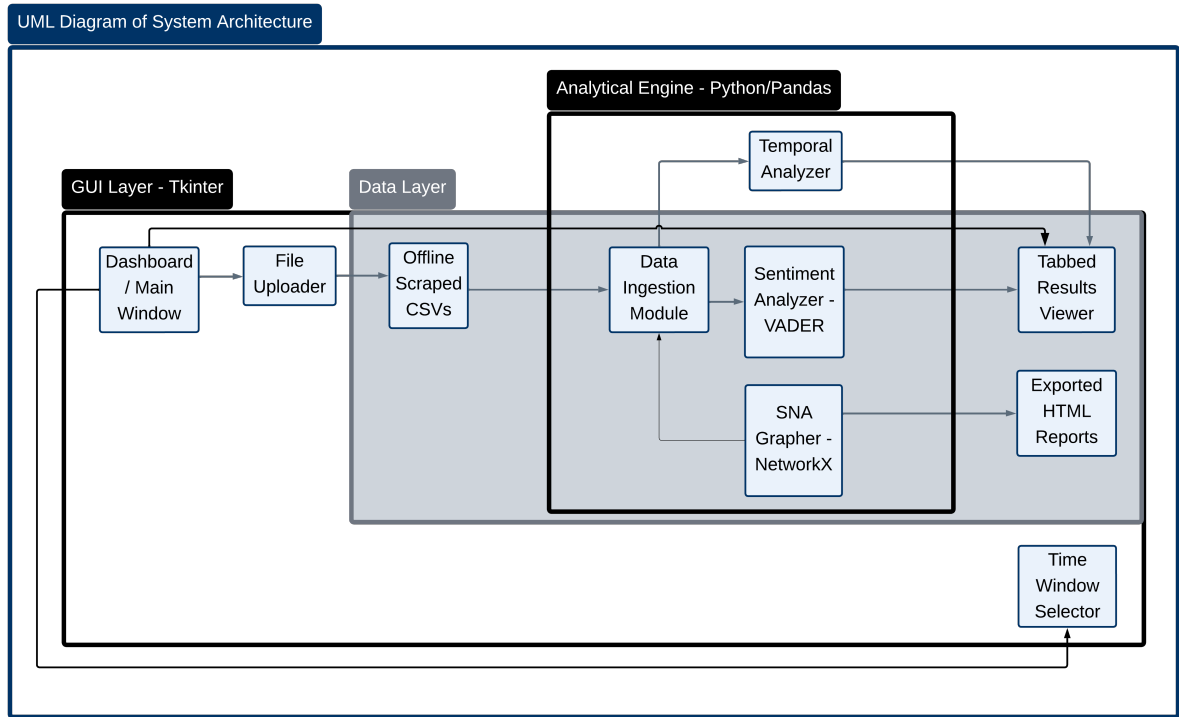


Figure 3.1: UML Diagram of System Architecture

This system operates on a post-acquisition premise, ingesting locally stored datasets (CSV/XLSX formats) representing the target's digital footprint. To uphold stringent ethical constraints and OPSEC, the system does not make outbound network requests to social media APIs. Furthermore, location data and time zone inferences are processed entirely offline using local geospatial databases (reverse_geocoder and timezonefinder). This methodological safeguard guarantees that sensitive coordinate data is never leaked to third-party geocoding services during the normalization phase. The acquisition layer will return a list of posts, which at minimum will include an exclusive post identifier, a timestamp, caption text (where provided), and an image file or image reference if the post includes media. The methodology assumes that the information in the posts will

be incomplete and that the pipeline will support null or missing fields. This reflects the reality of the OSINT situation, where publicly available information is heterogeneous and inconsistent.

Once collected, the system ingests the posted materials into memory Pandas DataFrames. To ensure persistence, the system generates standardized CSV exports of the normalized data. It separates short-run acquisition from long-run analysis, so an analytical run can be repeated using the offline CSV datasets without re-accessing the volatile data source. It also enables systematic evaluation of the pipeline, as intermediate findings can be validated and linked to crude notes. In a sense, the standardized CSV exports serve as an audit mechanism for traceability, a significant attribute of any intelligence workflow where outputs are expected to be traceable to their inputs.

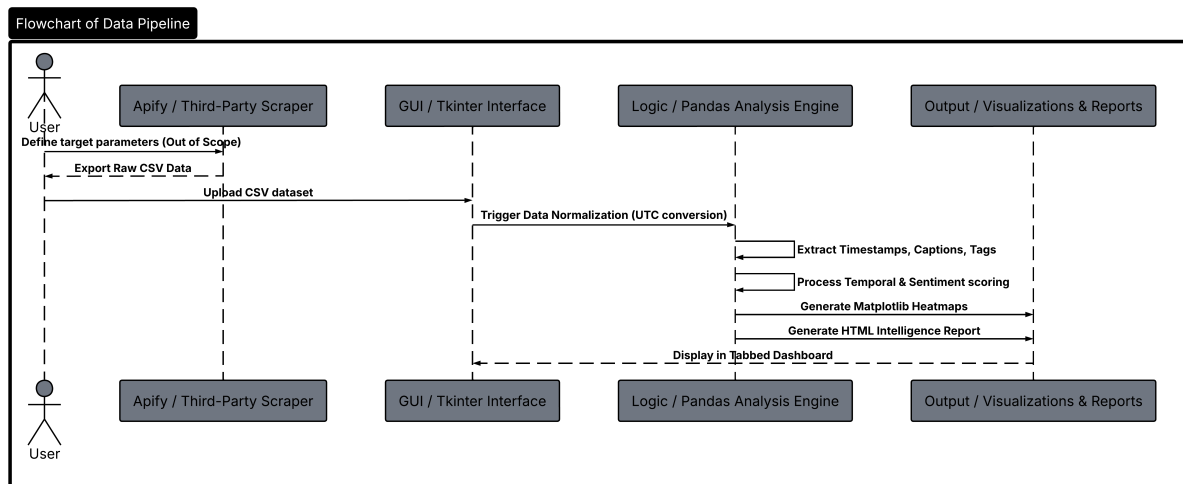


Figure 3.2: Flowchart of Data Pipeline

The user can interactively access the acquisition phase via the dashboard. The user selects a target account from a preloaded list, then selects a time window. The pipeline then fetches all posts, and time-window filtering is a deterministic transformation. This list is not random: it is better to first gather and then filter, as this provides coverage and eliminates bias introduced by biased retrieval, while the focus of analyses is not restricted to any time periods.

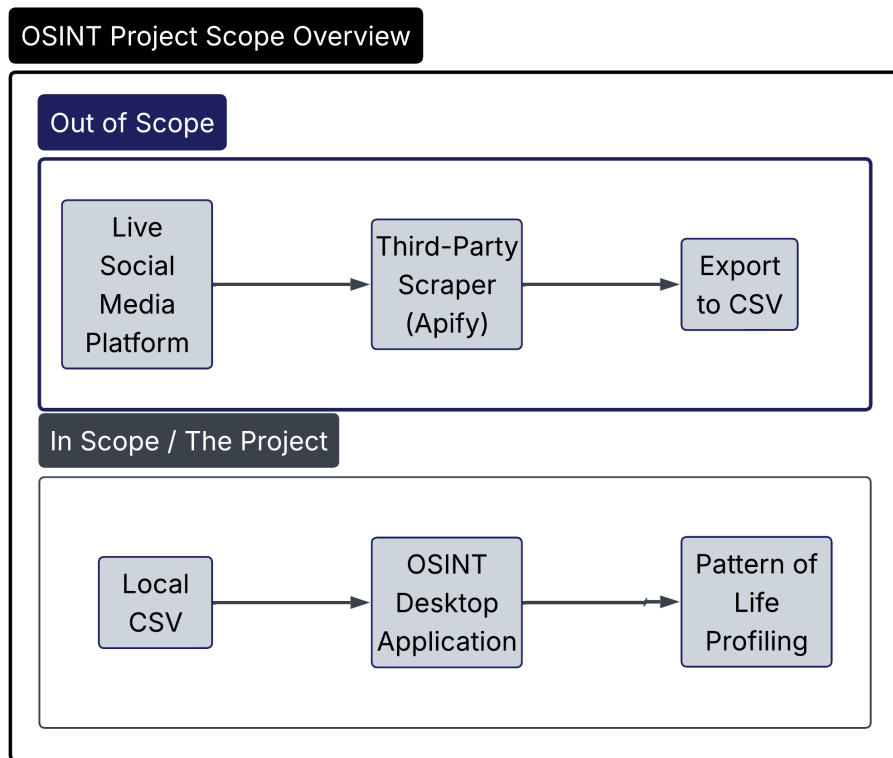


Figure 3.3: Out of Scope: Dataset generation, In Scope: Ingestion and Profiling

3.3.1 Synthetic Dataset Design Parameters

To ensure the system could be evaluated without violating GDPR constraints or ethical guidelines, the architectural design required the conceptualization of a fully controlled, synthetic dataset. Relying on live, arbitrary profiles would introduce unpredictable variables that could skew the validation of specific analytical modules. Therefore, six distinct target profiles were designed to systematically isolate key behavioral and technical variables.

The design parameters were explicitly mapped to the system's core capabilities to ensure comprehensive testing:

- **Media Origins (EXIF Validation):** Accounts were designed to simulate media uploads originating from both native device cameras and the platform's in-app camera. This parameter is critical for testing the system's Image Forensics module and validating platform-level metadata stripping.
- **Privacy and Relational Boundaries:** Profiles were divided into 'Public' and 'Private' configurations, utilizing varying frequencies of user tags and geotags. This tests the Social Network Analysis (SNA) engine's ability to map secondary leakage and spatial Pattern of Life (PoL) indicators even when primary access is restricted.
- **Sentiment Baselines:** To evaluate the sensitivity and accuracy of the VADER sentiment engine, target accounts were assigned specific emotional baselines (e.g., consistently positive, strictly negative, or highly volatile). This allows the system to be tested on its ability to detect programmatic 'mood drops' and emotional anomalies over time.

By defining these parameters during the design phase, the resulting dataset guarantees that every module—from spatial mapping to emotional profiling—can be tested against a known ground truth. The technical execution and specific data distribution of these synthetic accounts are detailed further in the Implementation phase (Section 4.2.3).

3.4 Temporal Windowing and Filtering Strategy

Another constraint of the methodological approach to PoL analysis is the inability to define a standard observation window. Without temporal scoping, frequency-based statistics and trend visualization cannot be connected, leading to the confounding of behavioral regimes and loss of anomalies. This system, therefore, renders temporal windowing a first-class concept, existing as predetermined possibilities and personal range capabilities.

The time window can be selected from three key modes: full-availability mode, which uses all retrieved posts; a mode with a rolling number of days, e.g., the last seven days or the last thirty days; and a mode with definite start and end dates. The method ensures that date inputs are in the correct format and order and displays error messages to the user whenever validation errors occur. Not usability-driven validation, but the hygiene of methodology. Instead, it can be used to ensure that the analysis results are linked to the intended temporal variables, which can be misinterpreted if left unchecked.

Once the user selects a time window, the system filters the dataset and performs computations exclusively on that chronological subset. This system captures a human-readable name for the window's actual boundaries and displays it on the dashboard. It is an interpretability property: it provides an anchor for every analysis result to a well-known period of observation, which is essential for credible reporting and reproducible results in the future.

3.5 Data Normalization and Analytical Representation

The system converts filtered posts into a tabular format. The conversion function transforms a list of post records into a data frame containing created fields for temporal analytics: date, hour of day, and weekday. This transformation has methodological implications because it clarifies the operational meaning of temporal PoL measures. Posts per hour (as an example) can be computed by grouping on an hour field created, and posts per weekday can be computed by grouping on a categorical representation of the weekdays in order rather than lexical order, i.e., Monday through Sunday.

A data frame structure is used for several methodological purposes. It supports reproducible computing, with explicit group-by operations, a medium for extracting summary metrics with aggregate functions, and visualization with plotting libraries. Moreover, DataFrames provide an open interface for testing because they allow exploration of intermediate representations and verification of their accuracy. This openness aligns with the dissertation's objective to present all the concepts and inferences that can be followed.

3.6 Pattern of Life Operationalization Through Temporal Analytics

This project operates the Pattern of Life construct through time regularities observed in posting behavior. Although PoL is a spatial and behavioral phenomenon, a time analysis is conducted relative to the posting time, which serves as a behavioral proxy for routine, availability, and inactivity intervals. It therefore computes time distributions across a set of granularities to facilitate baseline modelling and the visibility of anomalies.

The system has three major temporal products. The time series of posts/day is initially computed to reveal the activity on calendar days. The representation aids in identifying long-lasting changes/ changes in direction towards posting behavior, gaps (which may indicate absence), and bursts (which may indicate events). Second, the posts are categorized by time of day to estimate the daily routine pattern. The forms of these distributions can indicate regular posting times and be used to draw conclusions about day rhythms. Third, the posts of the weekday additive are designed to display the use of a weekly periodicity and the contrasting behavior between working-week behavior and weekend behavior.

The dashboard displays these time-based outputs under the temporal patterns tab. The methodological intention does not simply imply computing these statistics but also making them understandable to a non-expert user. This is important to methodology because PoL analysis cannot be validated unless the visual outputs of the analytical process can be deciphered, and the indistinctness of the axes undermines the quality of the analysis's findings.

The calculation of summary statistics is also intended to provide brief descriptions of the temporal profile. These are the total number of posts in the selected window, the number of days covered, the average number of posts per day, the median number of posts per day, the busiest hour, and the number of posts per hour. The measures facilitate the comparative analysis between accounts and windows and provide expedient pointers of routine intensity and regularity.

3.7 Technique of Sentiment Analysis and Competitiveness

The methodology employs sentiment analysis as the behavioral and expressive component of the PoL construct. The methodological aim is to determine whether there are discernible emotional dispositions in publicly visible captions over time and whether there is any prospect of sharing these dispositions with temporal structures. The sentiment analyzer computes a compound sentiment score for all text-based posts and aggregates the scores daily to generate a trend line.

The approach assumes that sentiment outcomes should be expressed as probability predictions rather than the psychological reality state. This is a determining factor in academic honesty. Lexicon-based sentiment tools, e.g., VADER, are both situation-, sarcasm-, and domain-sensitive; they can provide useful hints but cannot serve as absolute emotion metrics. Accordingly, the system presents descriptive statistics, including the mean, median, minimum, and maximum compound scores, along with a graph of average daily sentiment. In a complete write-up, each synthetic target should also in-

clude a concise interpretive outline of the direction and stability of sentiment trends over the selected period. This interpretive layer provides the analyst with brief, contextual conclusions, preventing unexplained gaps in the behavioral profile.

Sentiment results are displayed as a plot and a table of new posts and their corresponding scores for each compound in the interface. This doubled representation makes it easier to use: the user can see which texts produced extreme values and can tell whether the model's output is believable given the content.

3.8 Image Leakage/Metadata Risk Analysis

The image leakage module provides a second risk analysis level in accordance with the spatial PoL and incidental disclosure. The methodology aims to discover leakage opportunistically and report it, rather than if image data stored in platform-hosted environments will retain EXIF data. The system analyzes on-device images and focuses on two general types of leakage: explicit and implicit. Explicit leakage involves GPS positioning metadata, and implicit leakage involves text that can be observed in the image.

While the primary data normalization and temporal analysis are strictly offline, the image leakage module features an optional, user-triggered capability to fetch missing images directly via their CDN (Content Delivery Network) URLs. This is the only phase within the pipeline where outbound network requests are permitted. For analysts operating under strict Operational Security (OPSEC) conditions, this feature can be bypassed by ensuring all required media is locally cached alongside the dataset prior to ingestion.

The leakage analyzer provides structured logs for all processed images, including the post identifier, image name, extracted graphics positions (when present), and extracted text snippets (when present). These records are combined into cumulative findings, such as the total number of images analyzed, the volume of images with GPS metadata, and the volume of images with distinguishable text. The importance of such counts as evaluation metrics is that they measure the frequency of the most common leakage channels in the dataset and enable comparisons between accounts or time points.

3.9 Graphical User Interface as an Interpretability Layer

The tool is written in a standard GUI library running in a graphical user interface on the desktop, which presents a workflow that mimics the way an intelligence analyst would handle their task, which is to: select a target, select an observation window, run an analysis, and conduct a systematic review of the results. The interface separates the results into tabs by analytical categories, including an overview, temporal patterns, sentiment, image leakage, and raw posts. This segmentation is a methodologically approachable method because it overlays the PoL dimensions and the underlying pipeline stages, so the user knows which outputs result from which analysis processes.

The overview tab provides a descriptive background, including the effective time window and headline statistics. It also provides a list of recent posts, cut short to be readable, that gives a qualitative summary of the dataset. A table called "temporal patterns" displays the key PoL temporal analytics in chart format. The sentiment tab combines a trend plot, descriptive statistics, and a table of rated posts. The image leakage tab pro-

vides a summary count, record/image details, and a workflow to inspect image leakage. Auditability: Raw post identifiers can be supported in their raw state (without analysis) via the raw posts tab.

The interface is not only a convenient feature from a methodological perspective. It is the principal channel through which the analysis is expressed and understood. The dissertation, therefore, regards graphitization and understanding as elements of design methodology. To fulfill the project's demonstrative and educational goals, the outputs must be easily interpretable by users without extensive specialist knowledge.

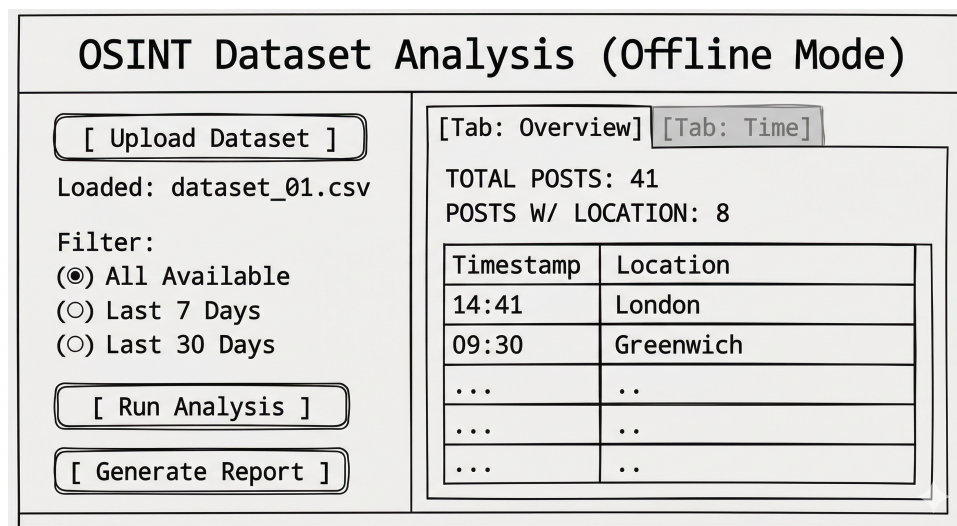


Figure 3.4: GUI Wireframe layout

3.10 Activities of evaluation and validation strategy

This is evaluated through a combination of functional validation, quantitative measures, and controlled-scenario analysis on a synthetic account. Functional validation ensures that each module produces results commensurate with its requirements. For example, to verify the validity of the aggregation results, the aggregated count is compared with the number of posts in the filtered dataset, and weekdays and hours are determined accordingly based on timestamps. Equally, the results of the sentiment are checked by the fact that a specified record of scored records is linked to a specific post, and that the result of the summation of the day depicts the average score of the underlying compound records.

System summaries are quantitative measures that are accessed. These temporal measures are the total number of posts, the number of days, the average number of posts per day, the time of day relative to the median, and the number of busy hours. The measures of sentiment include the mean, median, and spread of compound scores, as well as the stability and directionality of daily sentiment averages. The outcomes of the analysis of leakage metrics include the total number of images and the number of images with GPS or textual leakage. The data is artificial; therefore, additional verification is possible by establishing account behaviors known to exhibit patterns, e.g., consistent posting at a particular time or purposeful alteration of mood over time. The extent of reproduction of the outputs of the tool regarding the known patterns is an effect that acts as the known correctness checking.

PoL also employs scene-based assessment. Synthetic data is constructed to include routine-like behavior and at least some deliberate anomalies, such as sudden increases or decreases in posting frequency, incorrect geographic location information, or postings with captions that convey a mood. The methodology applied to the research questions to assess the presence of such events in the charts and summaries confirm the methodology's key argument: that the practice of aggregation reveals patterns and deviations even when the posts appear innocent.

3.11 Summary

Overall, the methodology implements an OSINT pipeline to present the privacy implications of aggregated social media traces in a clear, understandable way. It defines a managed data system in the form of synthetic accounts, achieves deterministic temporal filtering and aggregation to embody PoL routines, provides probabilistic sentiment analysis as a behavioral layer, and provides image leakage analysis to depict explicit and incidental disclosure routes. These outputs will be displayed in the system in a well-structured graphical dashboard that is friendly to audit and interpretation. The methodology provides support to the core thesis of the dissertation by virtue of functional and quantitative evaluation of the tool, controlled scenario testing, which holds that with aggregated and structured digital evidence of individuals, there are signs of patterns in life that may not be envisaged by the users at the post level.

Chapter 4

Implementation

4.1 Code Repository Outline

The full source code for this project is hosted on GitHub at: <https://github.com/Jorgearteaga83/PoL-Analysis-OSINT>.

The repository is structured as follows:

- /src: Contains the core Python scripts (ingestion, sentiment analysis), Tkinter interface code..
- /data: Directory for sample offline CSV datasets (synthetic data only).
- /output: Contains the generated intelligence reports and SNA graph outputs.

Table 4.1: System Modules & Technologies Mapping

System Component	Primary Function	Underlying Technology / Library
Data Normalization	Ingests offline, scraped data files CSV/XLSX, Cleanses missing values, parses metadata, and unify all timestamps to Coordinated Universal Time (UTC).	pandas
Temporal Engine	Calculates daily frequencies, active hours, and infers user time zones to establish a routine baseline.	pandas, timezonefinder
Geospatial & SNA	Extracts mentions, tags, and spatial entities to generate relational graphs using	reverse_geocoder, networkx
Sentiment Engine	Evaluates post captions using VADER (with heuristic fallback) to assign emotional polarity scores.	vaderSentiment, custom heuristic fallback
GUI & Visualization	Renders the interactive dashboard and dynamic heatmaps offline.	Tkinter, matplotlib
Image Forensics & Leakage	Extracts embedded GPS/EXIF metadata from local media and handles opportunistic URL image fetching.	Pillow (PIL), requests

4.2 Data Sources and Acquisition

4.2.1 File-Based Dataset Mode

The primary work of the system is the work that is associated with file-based datasets that are typically in the CSV format. The data will all be categorized based on social media communication (timestamps, comments, and media referrals, among any other metadata about the interaction). This type of analytics and assessment is the default one, as it can be reproduced under any conditions, and does not imply any ethical grayness regarding the use of live data. When one clicks on a dataset file in the user interface, the system will determine the format of input and retrieve the information in a form within the system. The different file structures are compressed into one schema to get the same analysis. In cases where no optional net worth is present, e.g., image filenames or location information, the system assigns safe default values to provide continuity in analysis. It is possible to evaluate such a system without having to use any external platform, and all the analysis

findings of any assessor can be fully duplicated by simply putting in the same input files through a data-first method.

4.2.2 Social Network Analysis (SNA) and Automated Reporting

Instead of relying on volatile live data access, the system extends its analytical depth through a Social Network Analysis (SNA) component. Using the networkx library, the system parses tagged users and mentions across the dataset to dynamically generate an interaction graph. The target user acts as the central node, with edge weights proportionally indicating the frequency of interaction with secondary entities.

Furthermore, the system synthesizes these multidimensional analyses into an automated HTML Intelligence Report. This reporting engine utilizes heuristic NLP categorization to classify inferred locations into distinct spheres (e.g., Work, Home, Travel) based on caption lexicons, providing a structured, intelligence-style product ready for analyst review. It is important to emphasize that this heuristic NLP approach serves strictly as a proof of concept. The system relies on static keyword matching to categorize locations, which lacks true semantic understanding. However, it successfully demonstrates the potential of automated spatial profiling for this project's scope. In a production environment, this rudimentary module acts as a lightweight substitute for more advanced, context-aware Natural Language Processing models, such as Bidirectional Encoder Representations from Transformers (BERT) or Named Entity Recognition (NER) architectures like spaCy (Devlin et al., 2019[13]; Honnibal et al., 2020)[19].

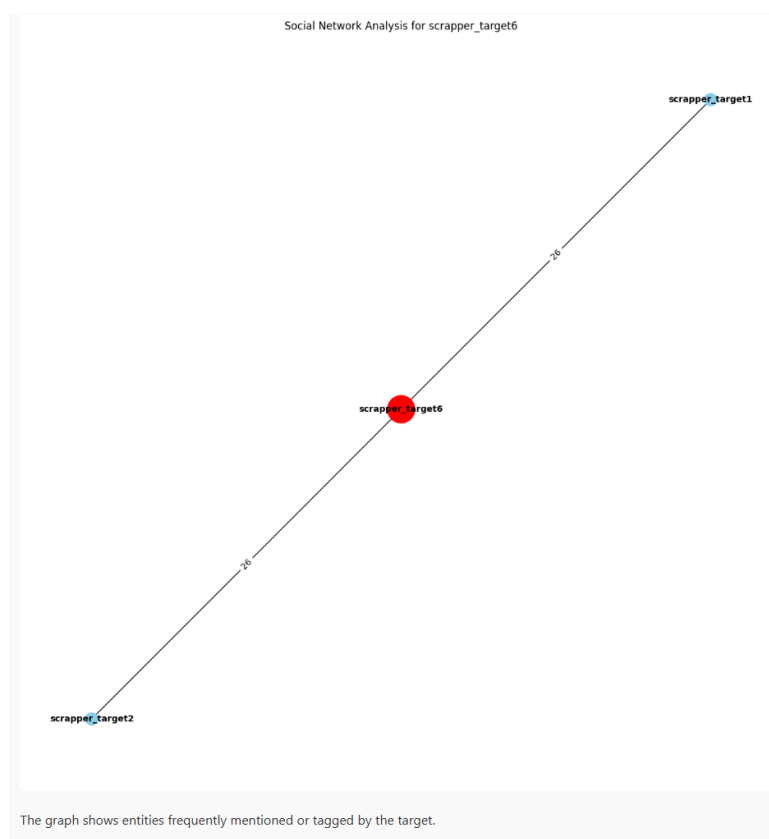


Figure 4.1: Social Network Analysis graph (SNA).

4.2.3 Synthetic Account Generation & Variable Control

To evaluate the OSINT pipeline against various privacy behaviors and data leakage vectors, six controlled synthetic Instagram accounts were developed. These accounts were engineered to isolate specific variables, including media source origins (testing EXIF data stripping), sentiment variability, and secondary leakage through tagging. The explicit specifications and analytical purposes for this controlled dataset are outlined in Table 4.2.

Table 4.2: Dissertation Data Analysis: Synthetic Account Specifications

Account ID	Privacy Setting	Camera Source	Tagging Behavior	Sentiment Baseline	Analytical Purpose
scraper_target1	Private	In-App Camera	No Tags	Negative	Baseline PoL and EXIF retention testing.
scraper_target2	Private	Device Camera	Places (Geotags)	Positive	Spatial PoL mapping and mood drop detection.
scraper_target3	Public	Mixed	2 * People (Tags) / 6 * Places (Geotags)	Negative	Testing Neutral and Negative sentiment.
scraper_target4	Public	Device Camera	3 * People (Tags) / 6 * Places (Geotags)	Positive	Testing Neutral and Positive sentiment.
scraper_target5	Public	In-App Camera	5 * People (Tags) / 8 * Places (Geotags)	Negative	Testing Neutral, Negative and positive sentiment.
scraper_target6	Public	Device Camera	4 * People, 1 * Location (Tags) / 8 * Places (Geotags)	Positive	Testing secondary leakage (SNA) behind privacy walls, Neutral, positive and 1 * Negative.

To provide a visual representation of these baseline configurations, Figure 4.2 displays the front-end layout of the six synthetic profiles.

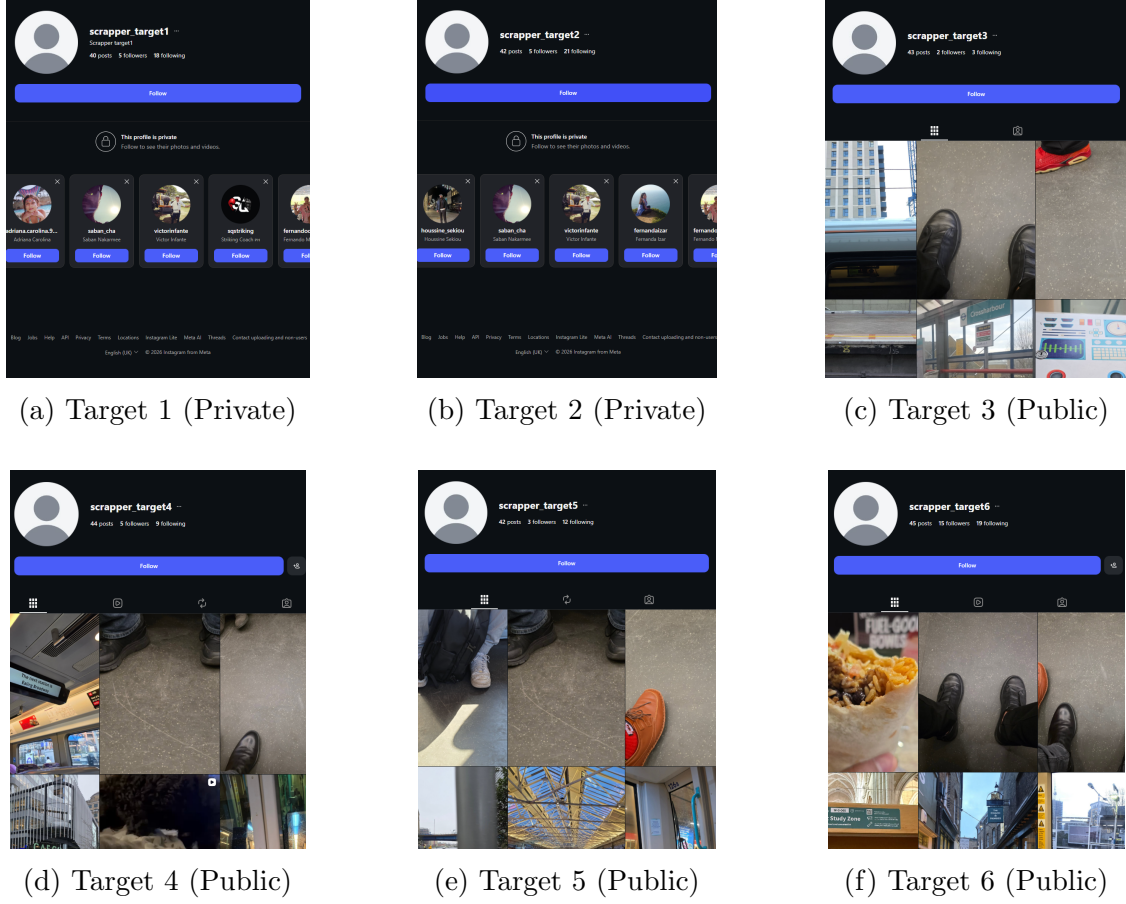
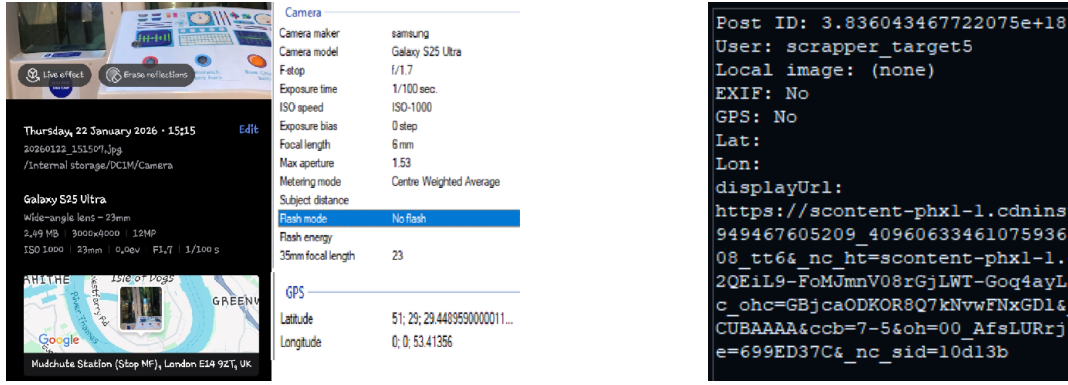


Figure 4.2: A view of the 6 synthetic Instagram profiles for the SOCMINT analysis.

A critical variable evaluated within this dataset was the persistence of embedded metadata. Target accounts were deliberately segmented to upload media captured directly via the native device camera (which natively embeds hardware headers and geospatial coordinates) versus media captured utilizing the platform’s in-app camera.

As illustrated in Figure 4.3, the platform employs EXIF data stripping upon upload. While the original device captured image retains metadata, including exact GPS coordinates, the resulting image hosted on the platform’s Content Delivery Network (CDN) is entirely sanitized of explicit location data. This visual evidence validates the system’s architectural decision to rely on heuristic text analysis and temporal inferences for spatial profiling, rather than depending on neutralized explicit EXIF harvesting.



(a) Raw EXIF data from device camera (GPS coordinates intact). (b) Extracted metadata from CDN image (EXIF stripped).

Figure 4.3: A technical comparison demonstrating platform metadata stripping. The raw image file (left) contains geospatial and device artifacts, whereas the platform-processed equivalent (right) has all identifying headers removed.

4.3 Data Preprocessing and Normalization

After acquiring the data, the system executes a preprocessing phase to normalize the datasets, ensuring analytical consistency and reliability. The algorithm converts all timestamps into a unified date and time format, adjusting them to Coordinated Universal Time (UTC). This will be a significant step into the temporal analysis because confusion, which is an outcome of the local time zones, will be done away with. The timestamp yields some values like the day of the week, the hour of the day, and the calendar date. Further analysis of the Pattern of Life is based on these attributes obtained.

Minor cleaning of textual items, e.g., post captions, does not lead to a loss of semantic meaning of the textual item. The system avoids aggressive text normalization (such as heavy stemming or stop-word removal) to preserve the original emotional context of the captions and the behavioral disposition. Null or empty captions are preserved in the dataset rather than discarded. In the context of behavioral profiling, the deliberate choice to post media without accompanying text is recognized as a distinct behavioral indicator in itself. The issues of implementation that will be addressed at the level are schema robustness. The areas of datasets from different sources cannot be identical. This schema harmonization framework equally arranges the rows of placeholders respectively in the circumstance that there is a need to avoid mistakes in lower analytics or originating.

Table 4.3: Standardized Data Schema

Normalized Field	Data Type	Analytical Purpose
timestamp_utc	Datetime (UTC)	Base metric for all temporal PoL analysis, day/hour aggregation, and routine mapping.
username	String	Acts as the primary key/filter for the target account under investigation.
location / category	String	Analyzed via heuristic NLP to classify spatial PoL (Home, Work, Travel).
associated_entities	List	Parses tagged users or mentions to populate the Social Network Analysis (SNA) graph.
caption	String	Ingested by the Sentiment Engine to calculate emotional tone and mood drops.
post_id	String	Primary identifier for individual digital traces; ensures traceability for auditability.
image_ref / display_url	String	Local path or CDN URL pointing to the post's visual media; utilized by the Image Leakage module.

4.4 Temporal Analysis & Pattern of Life Extraction

4.4.1 Temporal Windowing

The time filters operationalized in the GUI system include an overview of "All available" data, rolling observation windows for the "Last 7 days" or "Last 30 days", and specific custom date ranges. The system also actively validates inputs to ensure users provide properly formatted and logically sound custom date ranges.

In the instance of loading and normalization of data, it has been done, and thereby of the very acquisition of data, it is not temporal filtering that is involved. The design option will guarantee that the system will have deterministic characteristics, i.e., in cases when two or more of the parameters of the datasets and time window are identical, the system will give the same analysis outcome as the results of the analysis process. The interface effectively gives the time window in terms of visibility, hence rendering all further analysis to be anchored on a strict observation time.

4.4.2 Temporal Pattern Metrics

Three principal aggregations are obtained as Temporal Pattern of Life indicators:

- The number of posts made per day, frequency of posting, and non-use are revealed.
- Hour of day Posts, recording diurnal routine.
- Weekly publication in the arrangement of behavioral cycles each day of the week.

Then obtain from these aggregations, on the one hand, the summary statistics, including the mean and median number of posts per day, the hour of the day the most posts were made, and the number of posts made that particular hour. Together, these measures compose a collection of expected behavior of the analyzed account.

To drive quantitative and qualitative interpretation, numerical summaries are given, and temporal plots are given. The regularity (or periodicity) of posting behavior also plays a part in shaping the Pattern of Life baseline, and any breakage of the baseline can be one of the potential signs of abnormal occurrences.

4.5 Sentiment Analysis Implementation

The system applies sentiment analysis to post captions to track emotional tones and calculate their variations over time. To ensure software resiliency and continuous operation in constrained or offline environments, the sentiment module is engineered with a fault-tolerant, dual-layered architecture. Primarily, the system attempts to utilize the industry-standard vaderSentiment lexicon to compute compound sentiment scores bounded between -1 and 1, which efficiently discriminates between neutral, positive, and negative content.

However, recognizing that intelligence tools may frequently be deployed in air-gapped or dependency-restricted environments, a lightweight heuristic fallback lexicon is hard-coded into the SentimentEngine class. If the primary VADER library fails to initialize or is uninstalled, the system seamlessly degrades to this local heuristic model. This graceful degradation guarantees that the analyst can still extract baseline behavioral tendencies without the application crashing, allowing post-level scores to be saved and daily averages to be macro-inferred without interruption.

To take a case in point, the amount of feeling per day can be great, which can be explained by a personal or situational event.

The system is designed to reveal the results of the sentiments in descriptive statistics, the visualization on the time-series and the latest posts with the corresponding values in a tabular format. This is stratified presentation, transparent and gives the analyst an opportunity to determine the viability of the figures made by the text at the ground in terms of numerical sensations.

Table 4.4: Sentiment Engine Degradation Strategy

Execution Mode	Trigger Condition	Scoring Mechanism
Primary (VADER)	vaderSentiment library is successfully imported and initialized.	Lexicon and rule-based sentiment analysis returning a normalized compound score (-1.0 to +1.0).
Fallback (Heuristic)	vaderSentiment fails to load (e.g., strict air-gapped environment).	Lightweight, hardcoded dictionary comparing positive vs. negative word counts, normalized to approximate a -1.0 to +1.0 scale.

4.6 Pattern of Life Modelling

This pattern of Life analysis does not involve itself as a one-output of the algorithm but as an integrative reading of many dimensions of analysis. Pattern of Life is operationalized in three axes referred to as the temporal behavior, behavioral content, and the nature of the digital activities, which are the three attributes of the system.

Patterns of time dictate whether such a time is normal, the trend of sentiment offers a perspective into the emotional situation during time, and the pattern of media use offers an understanding of how such a subject interacts with the platform. A combination of these dimensions gives a behavioral baseline.

Violations in this reference frame are implicitly reviewed in the form of anomalies, e.g., missing a long time, posting out of regular working hours, sudden nature and mood, or a change in the content. This kind of intelligence analyst deviation is typically informative rather than stereotypical behavior, as they can point to travel, stress, change of operations, or concealment.

4.7 Graphical User Interface Design

The system is embodied in a self-contained desktop graphical user interface developed using the Tkinter library, structured around a tab-based system. To facilitate the interpretation of Pattern of Life routines, the interface directly embeds matplotlib to generate dynamic, offline temporal heatmaps and sentiment trend charts. The interface features a navigation control panel on the left and a main display area on the right, which organizes analytical results into logical, thematic tabs.

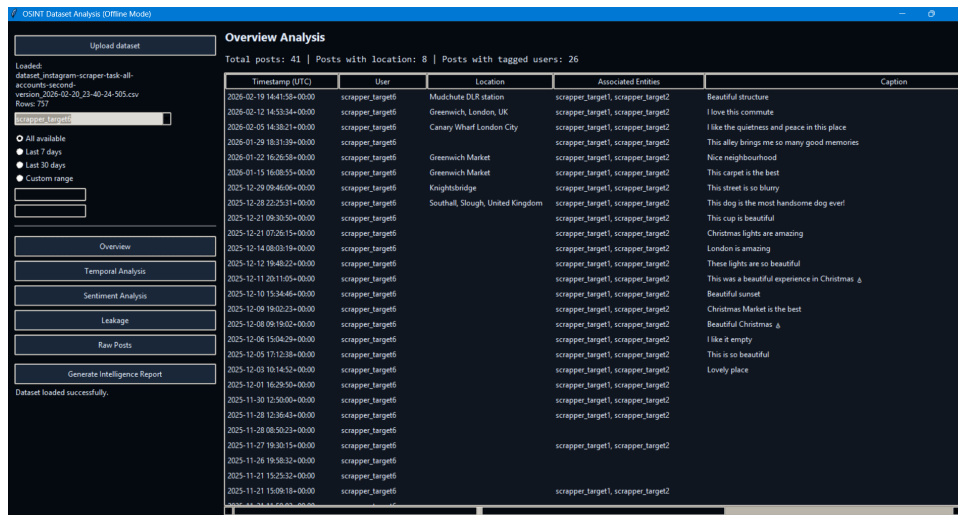


Figure 4.4: The primary OSINT analysis dashboard.

The workflow is set to work in a linear manner: select target, time window, source of data, parameters, and conduct analysis. Such a design can be used to reduce the cognitive load of the study and maintain the analysis to be controlled and repeatable.

Each analytical module includes certain tabs, including Overview, Temporal Patterns, Sentiment, Image Leakage, and Raw Posts. This split position allows the users to navigate freely between abstracts at high levels and details without their contextual de-orientation.

4.8 Reliability, Error Handling, and Robustness

Significant emphasis is placed on robust error handling and input validation during the implementation phase. Input validation prevents the usual traps of the users, such as out-of-range values or lost files. The runtime exceptions are caught with the help of descriptive messages and reported rather than allowing the application to terminate abruptly.

The particular focus is on the work with diverse datasets. Optional fields are handled defensively (thus, no data present does not have any influence on analysis or visualization). This approach presents actual OSINT conditions, where the completeness of the data cannot be ensured.

4.9 Chapter Summary

The chapter has presented a description of how Pattern of Life indicators can be extracted with the assistance of open-source data through an OSINT-based social media analysis system. It does so through modular architecture, effective preprocessing, and multi-dimensional analysis to demonstrate how behavioral intelligence may be concluded without having direct access to private information.

The next chapter evaluates the analytical results of the system, its strengths, and limitations, and considers the overall implications of the Pattern of Life analysis according to OSINT.

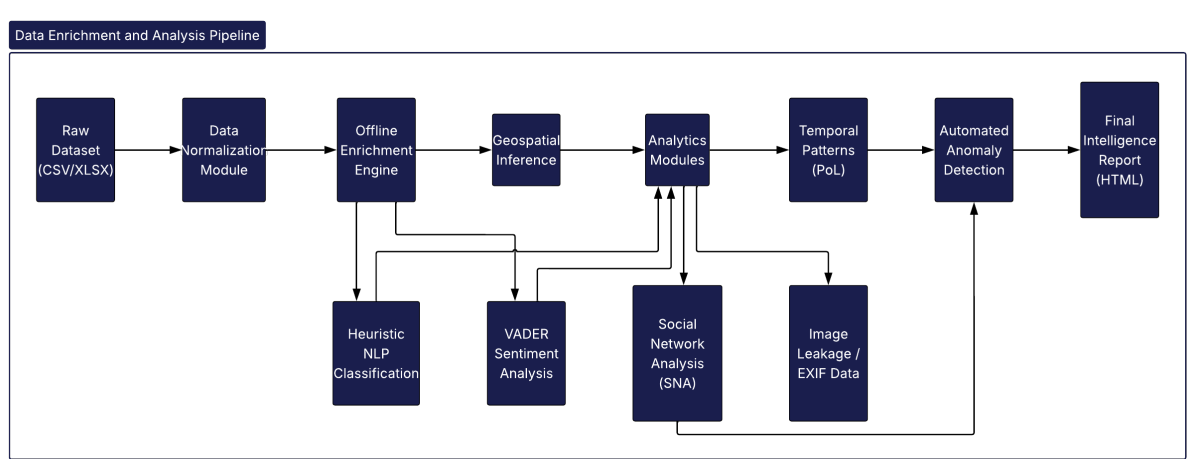


Figure 4.5: High-level data flow architecture.

The image above demonstrates the progression from offline data ingestion through heuristic enrichment, analytical processing, and final intelligence report generation.

Chapter 5

Results and Evaluation

5.1 Results

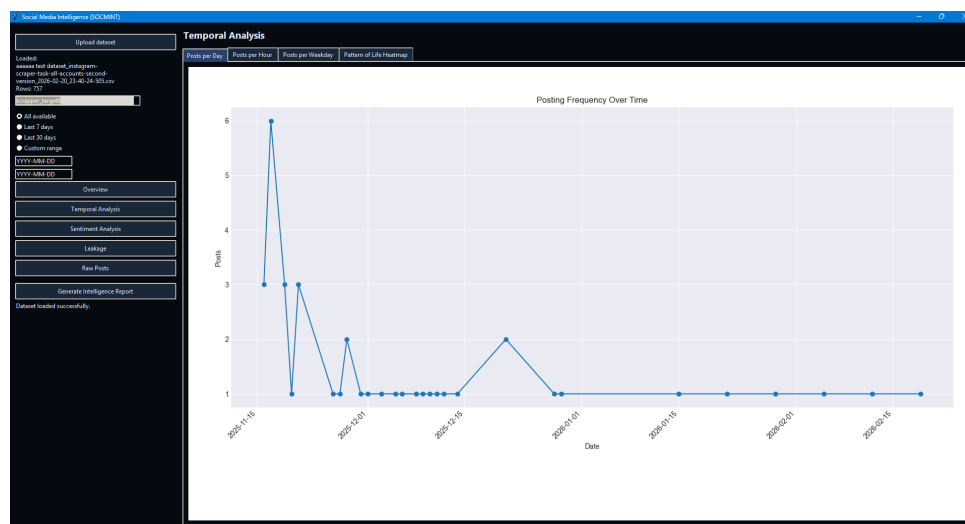


Figure 5.1: Temporal Analysis per day tab.

Temporal reports can be found by day showing the intensity of use for the period required.

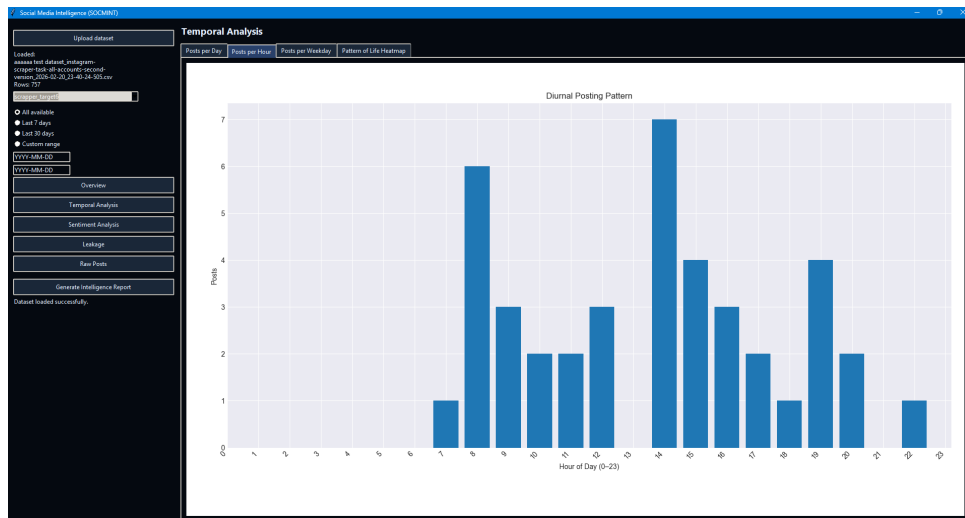


Figure 5.2: Temporal Analysis per hour.

Showing the most active hours of the user are 8 and 14.

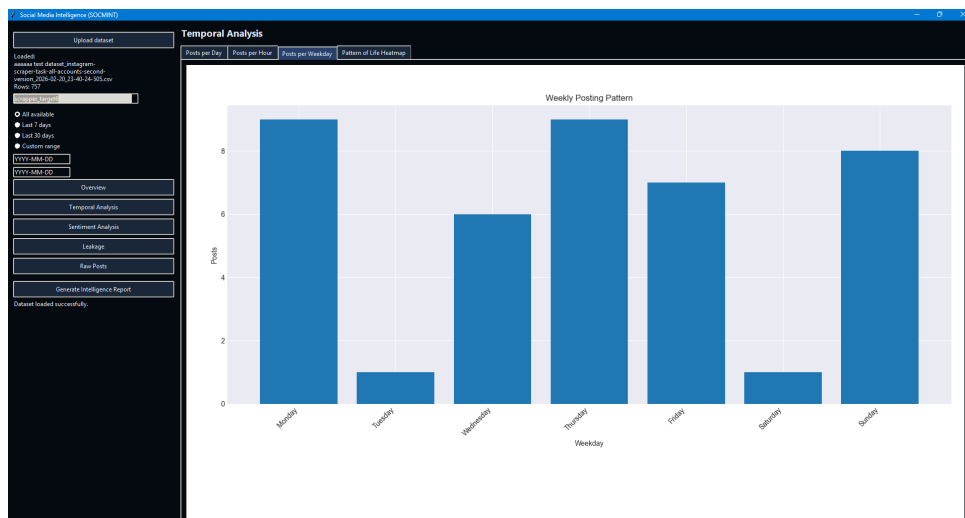


Figure 5.3: Post per weekday.

Showing Monday and Thursday as the most active days for the user.

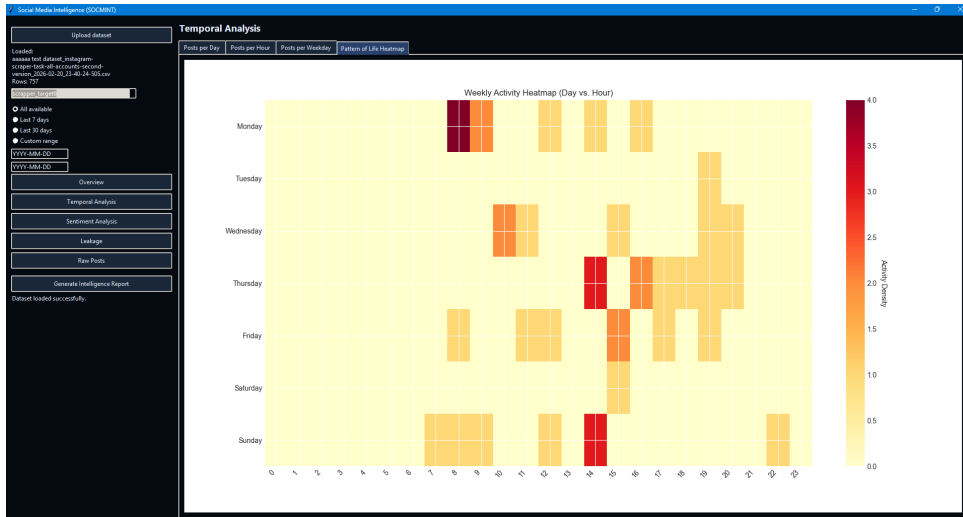


Figure 5.4: Pattern of Life heatmap.

Illustrating concentrated synthetic user activity at 8 on Monday and 14 on Thursdays, effectively establishing a behavioral baseline.

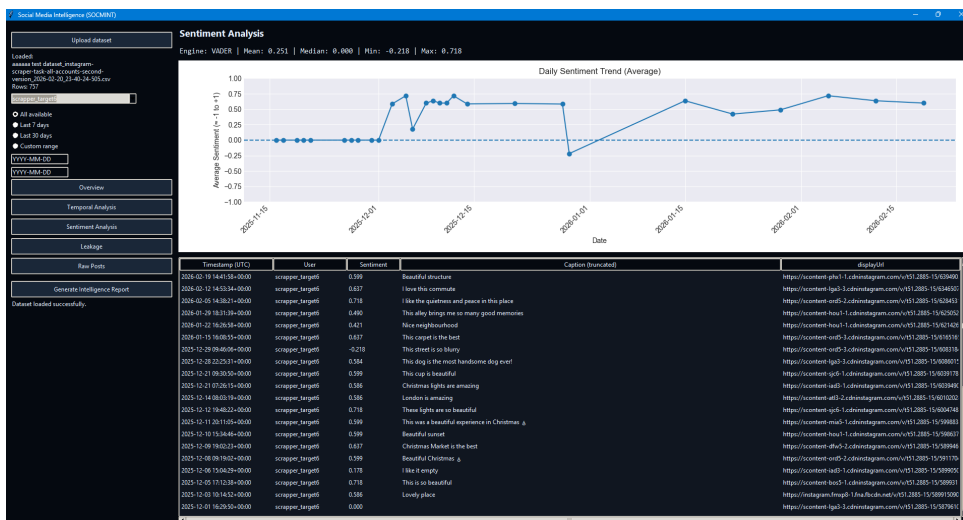


Figure 5.5: Daily sentiment trend aggregation.

The visualization smooths individual post noise to reveal macro-level emotional shifts over the observation window (2025-12-29, Sentiment: -0.218).

5.2 Discussion of Analytical Results

5.2.1 Temporal Behavior and Routine Identification

The temporal analysis reveals that even small datasets can portray consistent behavioral patterns when subjected to systematic OSINT aggregation. The results of the posting activity were observed to have certain clear patterns in terms of the timing of the day and the day of the week among the synthetic accounts investigated. These trends formed a good point of expectation of behavior that is among the conditions of the Pattern of Life analysis.

The posts about the hours and activity per week are the most used, they are the confirmation of the fact that the application of social media can be seen to indicate the pattern of an entire lifestyle (Cohen et al 2024)[9]. The posting activity in some cases was focused on some hours during the evening, which showed habitual use and not impromptu use.

This recurrence aligns with conventional intelligence analysis principles, which dictate that habitual behaviors are often more revealing than isolated incidents (Frasca et al., 2024)[14].

It is important to note that the period of inactivity in the system was as analytical as the period of activity. The set level was destabilized by long intervals between postings, and the potential behavioral changes were further emphasized. Although the artificial nature of the information does not allow one to apply the data to a real-life situation, the very mechanism demonstrates how temporal interference could be individualized into the form of an analytical signal.

5.2.2 Sentiment Trends and Behavioral Context

Sentiment analysis outcomes also added an additional interpretative layer that was complementary to time dynamics (Das, & Singh, 2023[10]; Photiadis & Papa, 2023)[36]. Although most of the posts were neutral in nature, there are positive or negative trends that could be monitored with a combination of frequency and timing of postings, these changes gave contextual details of the potential behavior or variations of the situation.

The sentiment visualizations on a daily average scale were particularly helpful to display the momentary emotional variations that could not be observed in comparison to the individual posts.

The effect of this method of aggregation was to reduce noise, whilst the representation of sentiment was an indicator of behavior and not a linguistic feature.

The results, though, also depicted the inherent limitations of the automated sentiment analysis, particularly when it is executed based on a short or formulaic caption. The large number of datasets had neutral scores, which showed that sentiment signals should be considered with care, and when used together with the other dimensions of analysis, and not separately.

5.2.3 Image Usage and Leakage Indicators

The image leakage evaluation yielded minimal findings regarding hidden GPS metadata. This accurately reflects modern platform operational security (OPSEC), as major social networks actively strip EXIF data including geolocation coordinates, camera models, and original timestamps during the upload and compression process. Consequently, extracting spatial information directly from platform-hosted images is unlikely, reinforcing the necessity of the tool's heuristic text analysis and offline time zone inference. This reduced the quantity of data that can be extracted based on locations but proved a crucial analytical result: leakage is also informative.

Where the images it offered contained a textual element that was visible, it had a more tangible analytical value. Although the synthetic datasets contained small amounts of text bearing images, the system could show how such materials could be removed, cataloged, and made available to specific posts. This feature is one of the reasons why visual media can also disclose contextual data although no explicit metadata is provided.

An image preview mechanism was added to improve the interpretability of the product significantly. The system facilitated explainability and trust in analytical procedures to bridge the gap between abstract outputs of the analytics and physical evidence by allowing analysts to visually study images pertaining to evidence of leakage as determined by the system.

5.3 Pattern of Life Interpretation

The evaluation does not measure the system on a single analytical dimension; rather, it assesses the system's ability to synthesize multiple metrics into a cohesive Pattern of Life profile. It was possible to observe behavioral patterns in time use, sentiment trends, and media use, comparing and assessing them over time.

The results have made it evident that Pattern of Life analysis can be operationalized based on open-source social media data, and there is no necessity to use the invasive and proprietary data sources. The system models baseline periodic practices to ensure it can clearly detect deviations from the user's standard routine. Such deviations in comparison to absolute values became the analytically most important outputs.

In the design of the system, anomaly detection is robustly operationalized through an automated heuristic engine. Rather than relying solely on manual observation, the system programmatically synthesizes deviations across multiple dimensions. It actively flags 'Travel Anomalies' by calculating sequential shifts in offline time zone inferences, detects 'Routine Disruptions' by identifying posts made during local sleep hours (e.g., 00:00–06:00), and highlights 'Mood Drops' by contrasting acute negative sentiment scores against a rolling sentiment average. These automated indicators act as a powerful decision-support mechanism, augmenting the analyst's ability to interpret the mosaic effect without being overwhelmed by the raw volume of digital traces. This architecture is similar to what is being practiced in the real world of intelligence, because the interpretation by humans will always be in the middle range, and the automated tools will be used as decision support tools, and not decision makers.

Table 5.1: Table 11 Anomaly mapping to its trigger

Anomaly Type	Conceptual Definition	Programmatic Trigger Condition (Heuristic)
Travel Anomaly	Sudden shift in physical location/region.	<code>inferred_timezone \$(t_n)\$ $\neq$ inferred_timezone \$(t_{\{n-1\}})\$</code>
Routine Disruption	Activity occurring during expected dormant periods.	Post time stamp falls between 00:00 and 06:00 (Local Time).
Mood Drop	Sudden, acute shift toward negative sentiment.	Post sentiment < -0.5\$ AND rolling 5-post average > 0\$.

5.4 System Evaluation against Objectives

The system's effectiveness is evaluated against its MoSCoW requirements and core research questions (RQs).

MoSCoW Compliance: The tool successfully implements all "Must Have" requirements, including the Python 3.x environment, secure CSV ingestion, offline temporal analytics, and a functional Tkinter GUI. "Should Have" features, such as VADER sentiment analysis and dynamic matplotlib heatmaps, were seamlessly integrated. The "Could Have" SNA module using NetworkX was also successfully deployed to map secondary entities and generate HTML reports.

RQ1 (Data Availability): The tool confirmed that while platforms routinely strip explicit EXIF GPS data from live uploads, substantial spatial and temporal metadata remains extractable via third-party scrapers into offline datasets, providing sufficient raw material for analysis.

RQ2 (PoL Reconstruction): The temporal engine successfully reconstructed reliable routines (e.g., active vs. dormant hours, workweek patterns) using solely aggregated UTC timestamps, validating that human behavior can be mapped without live tracking.

RQ3 (Sentiment Profiling): The VADER integration demonstrated that macro-level emotional shifts can be reliably plotted over time, acting as a behavioral barometer that provides contextual depth to the structural PoL data.

RQ4 (Risk Perception): By visualizing incidental leakage and routine predictability via the offline GUI, the tool tangibly highlights the severe privacy risks of continuous digital exhaust, successfully serving as an effective educational red-team artifact.

Chapter 6

Legal, Social, Ethical and Professional Issues

6.1 Introduction

Operating an OSINT tool requires strict adherence to legal and ethical frameworks.

6.2 Data Sources, Target Accounts, and Ethical Constraints

It is evident that the system is geared toward a small number of fake social media accounts created under this project's direction. The platform and the observer, referred to as the scraper or observer account, are used, while the rest of the accounts are regarded as targets. This is planned for target accounts with a time-based routine, such as posting time, caption text, media type, and the alleged location. The implementation of synthetic identities is not only an ethical safeguard but a standard requirement for robust operational security (OPSEC). Effective identity management and governance practices are critical when distinguishing between authentic and synthetic entities within complex digital ecosystems (Ghadge, 2024)[15]. By adhering to strict identity governance principles, the project ensures that the analytical tool remains focused purely on heuristic evaluation without inadvertently exposing or compromising real-world digital identities. The use of synthetic accounts provides two primary methodological advantages. Firstly, it eliminates the risk of physical harm to real participants and eliminates the need to tamper with the personal data of non-subjects. Second, it permits control of the variables of interest in PoL analysis through manipulations of experiments in their intended time regularity, location, and sentiment. This provides a yardstick to ensure the tool effectively measures or visualizes the desired patterns.

Synthetic design. In terms of synthetic design, the dissertation's emphasis on ethical vows also focuses on data minimization and controlled deletion. The system retains collected material only until evaluation and supports a workflow in which project data can be erased after analysis. Accordingly, the methodology treats the tool as an internal demonstrator rather than a product intended for distribution. This distinction is important, as the technical feasibility of OSINT aggregation does not necessarily imply the validity of its broad-based application.

6.3 Methodological Safeguards

The methodology has put in place precautionary measures to ensure the outputs are reliable and the constraints are well articulated. Samples Mixed period does not falsely interpret outputs of varying periods because of time-window choice and labelling. Traceability and repeat analysis. Data persistence helps prevent duplicate scraping and enables reproducible analysis. The modular design will ensure that a single point failure will not contaminate the rest of the pipeline, allowing the researcher to isolate defects. Raw posts and per-item statistics are displayed in the interface, making them interpretable and reducing the risk of overreliance on aggregate statistics.

Nevertheless, the methodology is also aware of the significant limitations of OSINT-based inference. Post-timestamp is a poor measure of offline behavior, as users can either post ex post or schedule their posts. Sentiment scores are crude and circumstantial and must be seen as indicators rather than diagnoses. Image metadata is not guaranteed to be available on every platform, and this stripping can occur; for example, spatial PoL inference with EXIF is not guaranteed to be opportunistic. Compression-sensitive image recognition of text may be resolution-, font-, and light-sensitive. Such disadvantages do not remove the demonstrator; only the number of claims that the dissertation can present is reduced. The harm in the method's contribution does not lie in the allegation that PoL inference is correct all the time, but rather in the possibility that actionable patterns and perceived exposure can be present when the conditions are not liberal.

6.4 Ethical Reflections and Legal Compliance (UK DPA 2018 / GDPR)

A primary strength of this project is its strict adherence to legal frameworks, specifically the UK Data Protection Act 2018 (DPA) and the UK General Data Protection Regulation (UK GDPR). Under the GDPR, processing publicly available personal data still requires a lawful basis and necessitates compliance with data minimization, purpose limitation, and the data subject's right to be informed. Scraping, storing, and analyzing live personal data of unaware citizens for an academic PoL tool would likely constitute a breach of these regulations, as the subjects have not provided explicit consent for their data to be aggregated into behavioral intelligence profiles.

To guarantee legal compliance and ethical integrity, this project exclusively utilized synthetic target accounts and artificially generated CSV datasets. By simulating realistic digital traces rather than harvesting real personal data, the project completely avoided processing Personally Identifiable Information (PII). This methodological safeguard allowed for rigorous testing of the tool's software architecture—proving the technical feasibility of OSINT aggregation—without violating the DPA or exposing non-consenting individuals to privacy risks. The system functions as a transparent, explainable decision-support tool, demonstrating the privacy vulnerabilities of the mosaic effect while upholding the highest standards of digital law.

6.4.1 Limitations of the Study

The project has several limitations despite the contribution it makes. Relying on synthetic data restricts the generalizability of the results, even though this approach remains an ethical necessity. Simulated datasets cannot be realistic because the behavior of social media in practice is richer, less predictable, and more situational.

The lexicon, text methods, and restrictions of the short format are the parameters that describe the sentiment analysis part. Less obvious emotional or contextual information may not happen to be effectively communicated without more advanced natural language processing techniques being employed.

Image metadata was not real, and it was also a limitation to image leakage analysis. Though this is a realistic behavior of platforms, it restricted the space where the analysis would be displayed.

While the system successfully implements heuristic automated anomaly detection, it stops short of applying deep predictive machine learning modelling to forecast future behavior. Future iterations could integrate predictive algorithms to anticipate user movements based on established baseline routines.

Furthermore, the system’s heuristic Natural Language Processing (NLP) for location categorization relies on static keyword matching mapped to three primary spheres: 'Work', 'Home', and 'Travel' (e.g., flagging 'office' as a Work location, or 'airport' as a Travel location). This rudimentary approach lacks contextual awareness and struggles with polysemy or idioms (e.g., 'working on myself' being incorrectly flagged as a workplace).

Chapter 7

Conclusion

7.1 Summary of Outcomes and Aims

This project successfully achieved its primary aim: developing, implementing, and validating a Python based OSINT application capable of ingesting, normalizing, and synthesizing pre-collected social media datasets. By utilizing a modular, offline analytical architecture, the system successfully translated theoretical OSINT concepts into a functional decision support tool. It generates composite Pattern of Life (PoL) and sentiment profiles, demonstrating intelligence value of publicly accessible digital traces.

7.2 Evaluation of Research Questions (RQs)

The outcomes of this research directly address and resolve the core Research Questions established on this study:

- **RQ1 (Data Availability):** The investigation confirmed that while major social platforms strip explicit EXIF GPS data from live uploads, temporal and textual data provided by users remains extractable. When processed into offline datasets, this data provides sufficient material for behavioral analysis.
- **RQ2 (Pattern of Life Reconstruction):** Regarding RQ2, the system proved that routine behaviors such as sleep cycles, active work hours, and weekend variations can be accurately estimated purely from temporal posting windows, without needing persistent live tracking or invasive surveillance.
- **RQ3 (Sentiment Profiling):** Through the integration of lexicon-based analysis, the project demonstrated that macro-level emotional shifts can be reliably plotted over time. This acts as a behavioral barometer, adding critical contextual depth to temporal routines.
- **RQ4 (Risk Perception):** By visualizing incidental leakage and routine predictability through a dedicated dashboard, the tool serves as an effective educational artifact. Highlighting the severe, yet heavily underestimated, privacy risks associated with everyday digital behavior.

7.3 Synthesis of the 'Mosaic Effect'

A central finding of this work is the validation of the "Mosaic Effect." The results show that even moderate or minimal levels of social media use can yield highly sensitive behavioral patterns when subjected to systematic OSINT aggregation. The inherent danger to user privacy does not lie in a single, seemingly benign post, but rather in the synthesis of data points over time. Time habits, sentiment, and incidental media leakage construct a detailed premise of behavioral deduction, bypassing the need for intrusive data sources or breached credentials.

7.4 Technical Limitations

Despite these successful outcomes, it is necessary to acknowledge the system's technical limitations to maintain academic objectivity. The pipeline's reliance on static keyword heuristics for Natural Language Processing (NLP) lacks deep semantic understanding, struggling with idioms or conversational context when categorizing spatial data. Furthermore, stringent API anti-scraping restrictions and aggressive metadata sanitization by platforms constrain the sheer volume of data available for automated collection.

7.5 Future Work

The system provides a robust foundation for future research and could be improved in several key areas. Future iterations could integrate advanced machine learning models to automatically identify behavioral deviations.

To overcome the limitations of static keyword heuristics, future iterations should integrate Named Entity Recognition (NER) models (such as spaCy) or Contextual Word Embeddings (like BERT). These models would allow the system to semantically distinguish between a physical location and a conversational idiom, reducing false positives in categorization. Additionally, future research could explore cross-platform Pattern of Life analysis, where data spanning multiple networks is correlated to build even higher-fidelity behavioral profiles.

7.6 Final Remarks

Ultimately, this project serves as a technical proof-of-concept that user awareness lags significantly behind modern OSINT capabilities. Driven by the "privacy paradox," users continuously exchange behavioral security for short-term social engagement, failing to account for how algorithms and analysts can aggregate their digital exhaust. This research highlights an urgent need for both platform-level privacy re-evaluations and enhanced user education regarding the compounding, permanent visibility of their digital footprints.

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Appendix A

Project Resources and Documentation

The following external resources, code repositories, and documentation have been made available to support the findings, implementation, and methodology detailed in this dissertation.

A.1 Project Proposal

The original project proposal outlining the initial scope, aims, and planned methodology for the OSINT tool.

Document Link: [Aggregating Unstructured Public Data for 'Pattern of Life' Analysis](#)

A.2 Source Code (SOCMINT Tool)

The complete Python source code for the SOCMINT OSINT tool, including the ingestion pipeline, offline temporal engine, sentiment analysis modules, and GUI architecture.

- **GitHub Repository (Primary):** <https://github.com/Jorgearteaga83/PoL-Analysis-OSINT>
- **Google Drive Backup:** [Source Code and Offline Assets](#)

A.3 Full System Documentation

Comprehensive technical documentation detailing the system architecture, module dependencies, installation instructions, and user guides.

Wiki Link: [DeepWiki Documentation: PoL-Analysis-OSINT](#)